

# A Multi-Objective Model for Traffic Signal Coordination Control With Queue Profile Estimation

Changze Li<sup>1</sup>, Yunxue Lu<sup>2</sup>, and Hao Wang<sup>3</sup>

**Abstract**—The research on signal coordination has been greatly enriched over the last decade. However, existing contributions face inherent limitations such as weak connection between objectives and common measurements of effectiveness (MOEs) caused by insufficient modeling of traffic dynamics, invariable phase splits, and great demand on hyperparameters. Meanwhile, nearly all related works are concentrated on scenarios with only under-saturated phases. Therefore, an arterial signal coordination model for minimum level of over-saturation and stops is proposed. Unlike most related works, the proposed model focuses on minimizing phase over-saturation and total stops by estimating queue profile for all phases under variable signal plans. The model is initially formulated as a mixed-integer nonlinear programming (MINLP). By applying linearization techniques, it is then transformed into a mixed-integer linear programming (MILP). Simulation experiments are carried out in SUMO, where an artery is built with eight scenarios of different traffic demand. The results indicate that the model is more competent in reducing average delay (AD), average stops (AS) and average total travel time (ATTT) than Yang's multi-path progression model for all scenarios. It is also verified to best MP-BAND by managing obvious reduction in AS and showing advantage in decreasing AD and ATTT in most scenarios. Additionally, the proposed model is able to alleviate the level of over-saturation for an intersection by re-allocating phase splits properly, resulting in less over-saturated phases. Intuitive illustrations attest to the effectiveness of the queue estimation in the proposed model, highlighting the theoretical importance of modeling queue length as a variable.

**Index Terms**—Traffic signal control, arterial signal coordination, minimum stops, over-saturated phases, queue profile estimation.

## I. INTRODUCTION

**T**RAFFIC signals play a crucial role in managing traffic flow at urban intersections. The coordination between signal plans is widely recognized as a valuable technique for enhancing traffic efficiency. Numerous models, strategies, and approaches have been developed to improve signal coordination by adjusting various parameters, including offsets, phase

splits, phase sequence, and cycle length. These studies aim to synchronize the signal plans across different intersections, enabling smoother and more efficient traffic movement.

In a general sense, the objectives of signal coordination can be classified into two categories. The first category aims to build a direct connection between the index of efficiency, such as delay and stops, and signal parameters [1], [2], [3], [4]. On the other hand, the second category focuses on maximizing bandwidth. Bandwidth-based models, compared to index-based methods, are often more concise and reliable. This is primarily due to the reduction of the complex traffic dynamics within their formulation, which is typically established as a mixed integer linear programming (MILP) model. The core idea of bandwidth-based methods is to maximize the continuous movement of traffic flow, enabling vehicles to pass through more intersections with fewer stops.

Signal coordination based on bandwidth is firstly proposed by Morgan and Little [5]. Their work was later revised by Little [6] and was enabled to proffer concurrent optimization of cycle length, offsets and speed. The method was eventually refined by Little et al. [7] as the well-known MAXBAND, which was built as a tool for two-way progression for through traffic with uniform bandwidth along each direction. MAXBAND was then integrated into many contributions with different progression focus. To account for scenarios where there are imbalanced two-way traffic volumes, a model was introduced by Lu et al. [8], which could produce more suitable plans than traditional methods did. Zhang et al. [9] proposed MaxBandLA and MaxBandGN, which were respectively dedicated to progression for long arteries and grid networks. Zhou et al. [10] made consideration for minor intersections in terms of geography and traffic volume and provided a method that could generate plans with uneven double cycling (UDC). Chen et al. [11], [12] developed a model integrating MAXBAND into diamond intersections with cross-over spacing optimization. In a more recent work [13], a model was presented which was capable of generating progression for multiple intersecting arteries in a grid network and a 3D time-space diagram was used for demonstration.

As aforementioned, MAXBAND enforces bandwidth uniformity along each direction, meaning that, for each way, the bandwidth is invariable between different intersections. In order to further unlock the potential of bandwidth-based models, Gartner et al. [14] proposed MULTIBAND, where the constraints on uniform bandwidth are removed so that

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bandwidths between two consecutive intersections are allowed to vary, making allowance for the change of traffic volume at different arterial parts. Several improved versions were also put forward by Gartner and Stamatiadis [15], [16] for network-level and route-based progression. The constraints in MULTIBAND which demand symmetrical bandwidth along the center-line are relaxed by Zhang et al. [17], who proposed AM-BAND which produced more flexible signal plans. A model named STREAM was presented by Yao et al. [18], which generated two-way progression with only sampled trajectories from connected vehicles (CVs). Nonetheless, its performance was still limited compared with MULTIBAND even if the penetration rate reached 100%. As MAXBAND, MULTIBAND and most of their derivatives are built on NEMA ring-barrier phase pattern, Jing et al. [19] offered a two-way progression model named Pband that could simultaneously optimize phase pattern choice along with other common signal parameters. Researchers also took into consideration the benefit of progression for transportation system with multiple modes. Dai et al. [20] and Lin et al. [21] respectively presented models which could generate signal progression plans for systems of coexisting passenger cars and buses. In the case of PM-BAND [22], progression plans were produced for both passenger cars and transit vehicles. Kim et al. [23] considered the uncertainty of dwell time for transit vehicles, proffering a bandwidth-based model that considered the initial queues at an intersection.

However, all the contributions mentioned above are concentrated on the progression for two-way through traffic. Consequently, they may fail to make the utmost of the system potential as through and turning volume may be at the same level for some urban networks. To cope with such deficiency, efforts have been made to allow for scenarios where progression is expected for more than two-way through traffic flow. Efforts of this kind can be divided into two categories depending on their targets. The first is designed with pre-given paths. Yang et al. [24] proposed a multi-path arterial progression model, which was able to optimize the sequence of phase groups and offer progression for multiple predetermined paths. Arsava et al. [25] took the origin-destination (OD) information as input for a model termed OD-BAND, which could also provide multi-path progression plans. The model was then upgraded to OD-NETBAND [26] for network-level progression. Yan et al. [27] made advantage of collected trajectories in connected environments to identify the target paths in a grid network. They formulated a MILP for the solution of network-level multi-band signal coordination (NMBSC) problem and established an algorithm to accelerate the solving process for large scale networks. Li et al. [28] extended the phase pattern from NEMA ring-barrier rule to a set of customized signal structure, which may contain overlapping phases, split phases and even UDC phases. Li et al. [29] also made allowance for network scenarios with complicated geographical interrelated paths occupied by passenger cars or trams. The idea of adaptive control and multi-path progression was combined by Wang et al. [30], who built a model that would produce plans providing progression to several predetermined paths and adaptive control to other uncoordinated phases. Studies

of the second category is rather rare, if compared to the first kind, which would let the model decide which paths would be the beneficiaries of green-bands. Chen et al. [31] formulated a model which would try creating progression plans with only intersection turning flow. Plans produced by the model were the result of trade-off between all possible paths along the artery. Another effort, namely MP-BAND, was by Xu et al. [32], whose model automatically determined which local green-bands would be activated for phase pairs between adjacent intersections and, further, how local green-bands were concatenated as connected green-bands.

Nevertheless, due to the limited explainability of green-bands in improving traffic flow efficiency, several common limitations remain unresolved. To clarify these limitations and highlight the contributions of the proposed model, the characteristics of representative multi-path band-based studies are summarized in TABLE I, alongside those of the proposed model. The key problems in existing works that the proposed model aims to address are summarized as follows.

#### 1) Weak Connection Between Bandwidth-Sum and MOEs:

Most band-based models are formulated to maximize the weighted sum of bandwidths. However, this objective does not directly correspond to improvements in measures of effectiveness (MOEs) related to traffic efficiency. In other words, the solution that yields the highest weighted bandwidth sum does not necessarily result in the most efficient traffic performance, as measured by delay, stops, or other MOEs.

#### 2) Oversimplified Queue Estimation:

Most existing band-based models do not attempt to estimate queue formation at intersections. Instead, queue lengths are either treated as fixed inputs or entirely disregarded. As a result, the actual vehicle movements between intersections may differ significantly from those implied by solutions that prioritize the maximization of weighted bandwidth sum.

#### 3) Fixed Splits and Dismissal of Over-Saturation:

Most existing studies do not allow phase splits to vary during the optimization process. Instead, they rely on pre-determined splits, often derived from models designed for optimizing timing plans at isolated intersections under under-saturated traffic conditions. This limitation inherently restricts the applicability of such approaches, as they tend to overlook scenarios involving over-saturated phases. Consequently, critical aspects of traffic dynamics are largely neglected in the related literature.

#### 4) Burden of Hyperparameters:

It is important to recognize that most existing studies of band-based models are vulnerable to sub-optimal outcomes due to the substantial influence of hyperparameters, such as objective weights and minimum bandwidth, whose values are difficult to determine theoretically and often require costly field surveys or experiments to estimate reliably.

TABLE I  
A SUMMARY OF MODEL CHARACTERISTICS REGARDING TYPICAL MULTI-PATH BAND-BASED MODELS

| Model              | Objective                                              | Hyperparameters                                                                                                                                        | Decision variables                     | If over-saturated phases are considered |
|--------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|-----------------------------------------|
| Yang's Model [24]  | Sum of weighted path bandwidth                         | Cycle length, queue discharging time, minimum effective bandwidth                                                                                      | Phase sequences, offsets               | No                                      |
| OD-BAND [25]       |                                                        |                                                                                                                                                        |                                        |                                         |
| Yan's Model [27]   | Sum of weighted local and connected bandwidth          | Cycle length, minimum effective local bandwidth, minimum effective connected bandwidth, relative weights of different objectives, minimum phase splits | Phase sequences, offsets, phase splits |                                         |
| MP-BAND [32]       |                                                        | Cycle length, minimum effective local bandwidth, minimum effective connected bandwidth, relative weights of different objectives                       | Phase sequences, offsets               |                                         |
| Chen's Model [31]  | Sum of weighted local and effective bandwidth          | Cycle length, minimum effective bandwidth, relative weights of different objectives                                                                    |                                        |                                         |
| The proposed model | Level of over-saturation and total stops on the artery | Cycle length                                                                                                                                           | Phase sequences, offsets, phase splits | Yes                                     |

Building on the above discussion, the main contributions of the work are as follows:

- 1) An arterial signal coordination model is developed that simultaneously optimizes phase splits, offsets, and sequences, without relying on predetermined vehicle paths or hyperparameters beyond a common cycle length.
- 2) The model minimizes total over-saturation and vehicle stops by integrating local green bands with queue profile estimation, capturing both queue accumulation and dissipation.
- 3) The problem is first formulated as a mixed-integer nonlinear programming (MINLP) and then transformed into a MILP using relaxation techniques, including piecewise McCormick envelopes, to enhance tractability.
- 4) The proposed model is evaluated under varying traffic demand conditions to verify its applicability and effectiveness, with both quantitative and qualitative analyses provided to illustrate its performance and characteristics.

The remainder of the paper is organized as follows. **Section II** introduces the problem in detail. **Section III** presents the formulation of the proposed model. **Section IV** reports numerical examples and simulation results under various traffic conditions, with additional discussions. Finally, **Section V** concludes the paper and outlines directions for future research.

## II. PROBLEM DEFINITION

To address the limitations of traditional band-based models for arterial signal coordination, this study develops a

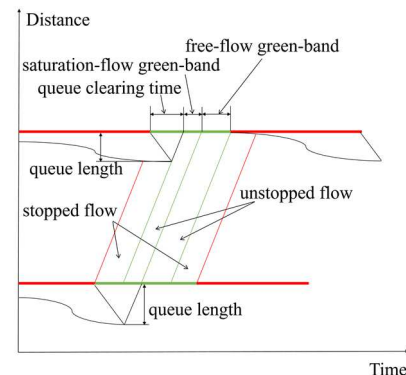


Fig. 1. Illustration for saturation-flow and free-flow green-bands and queue length.

framework that strengthens the link between green-bands and traffic flow by means of queue profile estimation. The queue profile for each phase consists of two parts—queue accumulation and queue dissipation—which together allow the estimation of queue length and clearing time. These two parameters are central to reallocating green time and to evaluating whether a phase is over-saturated or not.

First, the concept of local green-bands is redefined. Instead of a single uniform band, two distinct types are introduced: one representing saturation flow and the other representing free flow, as shown in Fig. 1. This differentiation provides a clear

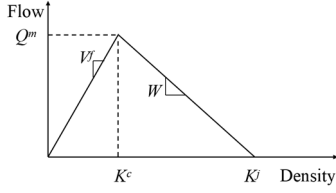


Fig. 2. Triangular fundamental diagram.

picture of how vehicles move between adjacent intersections. Importantly, vehicles traveling outside either type of green-band are assumed to be stopped and added to the downstream queue.

Second, with the aid of these differentiated green-bands, the process of queue accumulation can be estimated. Specifically, the downstream queue is formed by vehicles from the upstream intersection that are not accommodated by either green-band type. The resulting queue length represents both the vehicle stops and the basis for modeling the dissipation process. Building on this, the dissipation stage is described by calculating the time required to clear the queue (Fig. 1). This clearing time then serves as a direct indicator of phase performance: if the clearing time exceeds the green time allocated, the phase is over-saturated; if not, it remains within capacity. The difference between clearing time and allocated green time provides a quantitative measure of over-saturation. Additionally, the queue profile estimation establishes a direct link between adjacent intersections, eliminating the need for the additional connectivity formulations commonly required in previous studies.

Lastly, the queue profile estimation—together with the clearing times—provides a foundation for adjusting phase splits and other signal parameters. This adjustment process allows the model to search for solutions that minimize both phase over-saturation and vehicle stops. The tight connection between signal parameters and MOEs substantially reduces reliance on hyperparameters. In doing so, the framework yields an arterial coordination strategy that is theoretically explainable, explicitly linking the role of green-bands to measurable traffic efficiency outcomes.

The model is developed under a set of assumptions, which are introduced below to clarify its scope and applicability:

1) The triangular fundamental diagram (Fig. 2) functions as the basis of discussion on how queues accumulate and discharge. FD can be calibrated with capacity  $Q^m$ , free flow speed  $V^f$  and jam density  $K^j$ . Other parameters can be calculated such as critical density ( $K^c = Q^m/V^f$ ) and congested wave speed ( $W = Q^m/(K^j - K^c)$ );

2) In the proposed model, the traffic arrival is assumed to be stable and approximately uniform within traffic flows of different density, which will be revealed in model formulation;

3) The proposed model can work with the full knowledge of arterial OD information but is presented as the less demanding version, which makes use of intersection turning flow and a limited amount of collected trajectories from connected vehicles, to enhance its practicality;

TABLE II  
NOTATIONS FOR SUBSCRIPTS

| Notation | Definition                                                                                                                                      |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| $k$      | Subscript $k$ is used to indicate the intersection index in a variable while $k+1$ represents the downstream intersection of intersection $k$ . |
| $l$      | Subscript $l$ is used to indicate a signal phase at intersection $k$ in a variable while $l'$ is used for phases at downstream intersections.   |

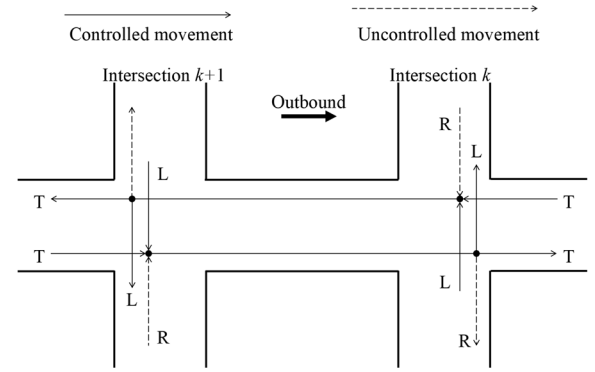


Fig. 3. Illustration on traffic movements.

4) The penetration rate of connected vehicles (CVs) is assumed to be uniform throughout the artery in order that the number of trajectories serves as an authentic reflection of the traffic flow between OD.

### III. MODEL FORMULATION

This section functions as a detailed introduction to the proposed model, which are comprised of six parts: the objective function, the saturation constraints, the local green-band constraints, the saturation-flow and free-flow green-band constraints, the stop estimation constraints and the phase sequence constraints. Definitions for subscripts in the model are given in TABLE II.

Building on the above definition, an example is given for better understanding:  $b_{l,l',k}^s(\overline{b_{l,l',k}^s})$  is the variable indicating the bandwidth of the local outbound (inbound) saturation-flow green-band from phase  $l$  at intersection  $k$  to phase  $l'$  at intersection  $k+1$  (s). To save room, explanations on subscripts would be dismissed if unnecessary in the rest of the study.

Prior to details of model formulation, it is indispensable to clarify that the local progression is built on the basis of traffic movements between adjacent intersections, which are shown in Fig. 3, where through and left-turning flow are under the regulation of signal plans while right-turning movement is deemed uncontrolled.

Further, phase pairs are introduced as the representation of movements. Thus, there are four kinds of phase pairs and they are through-through (TT), through-left (TL), left-through (LT) and left-left (LL). The first phase in a phase pair is called

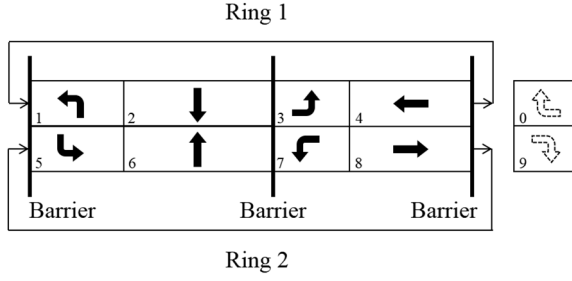


Fig. 4. The NEMA ring-barrier signal pattern.

an origin phase and the second is called a destination phase. Similar definitions apply to the inbound direction.

To make consideration for the effect of right-turning movement, a slight adaptation is made on the NEMA ring-barrier signal pattern as shown in Fig. 4, where two right-turning phases are named. Thus, the nomination of phase sets adopted in the model is given as (1)-(2).

$$\begin{aligned} L_k^o &= \{3, 6\}, L_k^d = \{1, 6\}, \bar{L}_k^o = \{2, 5\}, \bar{L}_k^d \\ &= \{2, 7\}, \forall k \in |K| \end{aligned} \quad (1)$$

$$\begin{aligned} L_k^u &= \{4, 8\}, L_k^r = \{0\}, \bar{L}_k^r = \{9\}, L_k \\ &= \{1, 2, 3, 4, 5, 6, 7, 8\}, \forall k \in |K| \end{aligned} \quad (2)$$

where  $L_k^o$  ( $\bar{L}_k^o$ ) is the set of outbound (inbound) origin phases;  $L_k^d$  ( $\bar{L}_k^d$ ) is the set of outbound (inbound) destination phases;  $L_k^r$  ( $\bar{L}_k^r$ ) is the set of outbound (inbound) right-turning phases;  $L_k^u$  is the set of phases which would not benefit from signal coordination as origins;  $L_k$  is the set of all controlled phases;  $K$  is the set of intersections on the artery and  $|K|$  is the cardinality of the set.

#### A. The Objective Function

Although band-based models have been extensively developed to maximize the weighted sum of bandwidths, their effectiveness is fundamentally constrained by limitations in objective formulation. In particular, these models often lack rigorous theoretical justification for the assumed correlation between increased bandwidth sums and overall traffic system efficiency. Furthermore, the bandwidth weights function as hyperparameters, typically selected empirically rather than derived through theoretical principles. Bandwidth-based approaches also exhibit limited utility in evaluating traffic conditions where green bands are absent, rendering them less effective in diverse scenarios. Additionally, the applicability of these models is further limited by their predominant focus on under-saturated traffic conditions, which diminishes their relevance and effectiveness in addressing congested scenarios.

Accordingly, the proposed model has two objectives: minimizing over-saturation across signal phases, which prevents queue growth and spill-back that degrade road capacity and safety, and minimizing vehicle stops, a key indicator of traffic efficiency. These objectives align with both safety and efficiency goals, eliminating the need for precise weight

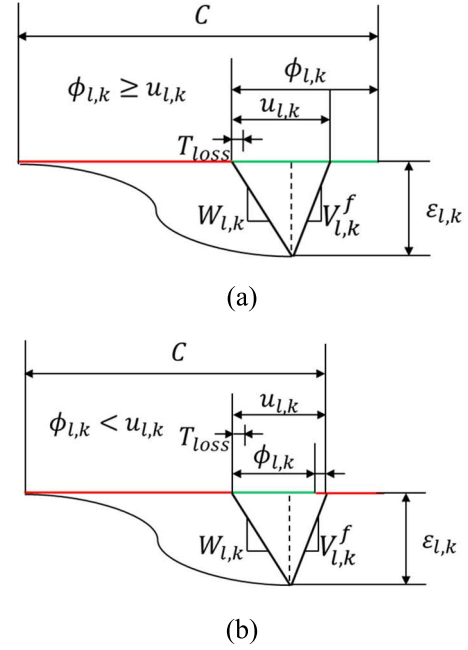


Fig. 5. Illustrations for (a) under-saturated phases (b) over-saturated phases. The accumulation of queue is simplified.

calibration, and enhancing the model's robustness and stability for real-world implementation.

$$\begin{aligned} \text{Min} \sum_{k \in K} \sum_{l \in L_k} \eta_{l,k} \\ \text{Min} \sum_{k \in K} \sum_{l \in L_k} \varepsilon_{l,k} \end{aligned}$$

where  $\eta_{l,k}$  is the variable indicating the level of over-saturation(s);  $\varepsilon_{l,k}$  is the variable indicating the queue length(m).

#### B. The Saturation Constraints

As a matter of fact, the level of saturation for a phase changes as splits changes. Thus, basic concepts regarding how phase splits can vary under NEMA ring-barrier rules should be given first. According to Fig. 4, fundamental constraints are set as (3)-(5). Constraint (3) indicates the sum of phase splits within the same ring should be equal to the common cycle. Constraints (4)-(5) are built for they describe how barriers in NEMA work. Also, a minimum green time constraint (6) is necessary considering engineering practice.

$$\phi_{1,k} + \phi_{2,k} + \phi_{3,k} + \phi_{4,k} + 4T_{loss} = C, \forall k \in |K| \quad (3)$$

$$\phi_{1,k} + \phi_{2,k} = \phi_{5,k} + \phi_{6,k}, \forall k \in |K| \quad (4)$$

$$\phi_{3,k} + \phi_{4,k} = \phi_{7,k} + \phi_{8,k}, \forall k \in |K| \quad (5)$$

$$\phi_{l,k} \geq g_{l,k}^{\min}, \forall k \in |K| \quad (6)$$

where  $\phi_{l,k}$  is the phase split variable(s);  $T_{loss}$  is the time loss for a phase(s);  $C$  is the predetermined common cycle length(s);  $g_{l,k}^{\min}$  is the minimum phase length(s).

One of the key features of the model is the intuitive definition of phase over-saturation. Illustrations of a full signal cycle can be seen in Fig. 5, where a simplified queue profile is

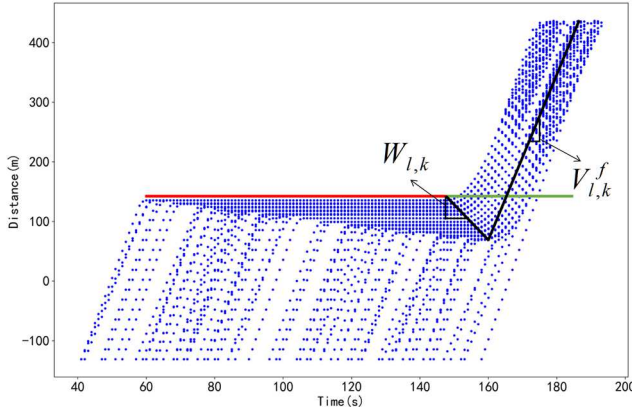


Fig. 6. The gaining of  $W_{l,k}$  and  $V_{l,k}^f$  from aggregated trajectories.

provided as a curve. Saturation flow  $W_{l,k}$  and free flow speed  $V_{l,k}^f$  are introduced to depict the accumulation and discharge of queue. Specifically, phases are thought to be under-saturated when the queue length can remain stable across multiple cycles, as shown in Fig. 5(a), where the phase split is no less than the queue discharging time including time loss  $T_{loss}$ , i.e.,  $\phi_{l,k} \geq u_{l,k}$ . Otherwise, over-saturated phases are inevitable because the available split is not long enough to fully discharge the queue, suggesting  $\phi_{l,k} < u_{l,k}$ , as shown in Fig. 5 (b).

In accordance with what is illustrated in Fig. 5, the queue discharging time for a phase is formulated as constraint (7).

$$u_{l,k} = \frac{\varepsilon_{l,k}}{W_{l,k}} + \frac{\varepsilon_{l,k}}{V_{l,k}^f} + T_{loss}, \forall l \in L_k, \forall k \in |K| \quad (7)$$

where  $u_{l,k}$  is the variable indicating the queue discharging time(s);  $W_{l,k}$  is the saturation flow(m/s);  $V_{l,k}^f$  is the free flow speed(m/s).

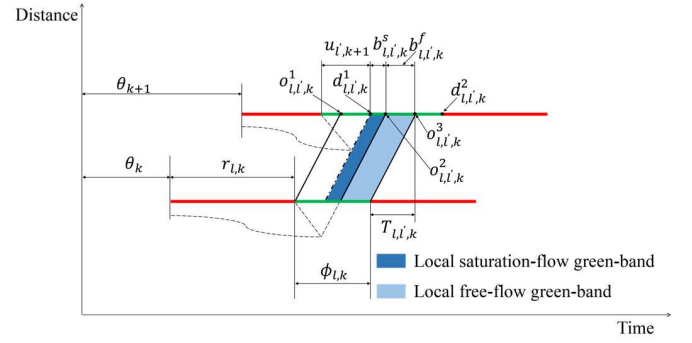
As the proposed model is designed for environments with low penetration rates,  $W_{l,k}$  and  $V_{l,k}^f$  are directly extracted from vehicle trajectories, which can be collected from different signal cycles and then aggregated within one cycle to obtain a reliable estimation, as shown in Fig. 6. Therefore, the demand on penetration rates is reduced to the minimum level. Moreover, with higher saturation levels and the presence of congestion, the achievable free-flow speed falls below the nominal value illustrated in Fig. 2. Accordingly, the value estimated through fitting to the trajectory points is used as the actual free-flow speed in the model.

As for phase saturation, constraint (8) is built with possible relaxation considering over-saturation, suggesting the constraint is only effective when phases are under-saturated.

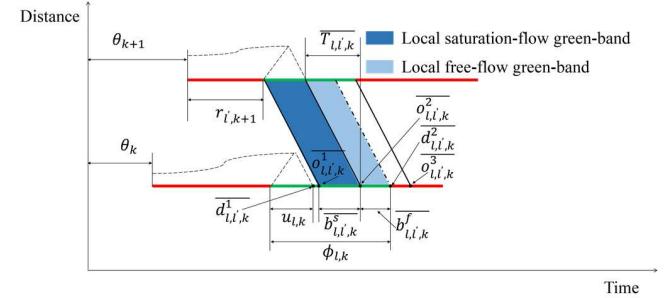
$$0 \leq u_{l,k} \leq \phi_{l,k} + \psi_{l,k}M, \forall l \in L_k, \forall k \in |K| \quad (8)$$

where  $\psi_{l,k}$  is the binary variable indicating whether a phase is over-saturated or not. The variable is equal to 1 if the phase is over-saturated;  $M$  is the big number for relaxation.

Based on the above discussion, how the level of over-saturation for a phase ( $\eta_{l,k}$ ) is calculated is shown in constraints (9)-(10). In (9),  $\eta_{l,k}$  is deemed equal to 0 if a phase is under-saturated. Constraint (10) suggests  $\eta_{l,k}$  is calculated as



(a)



(b)

Fig. 7. Notations for saturation-flow and free-flow green-bands considering queue. (a) outbound (b) inbound.

the difference between queue discharging times and phase splits.

$$|\eta_{l,k}| \leq \psi_{l,k}M, \forall l \in L_k, \forall k \in |K| \quad (9)$$

$$|\eta_{l,k} - u_{l,k} + \phi_{l,k}| \leq (1 - \psi_{l,k})M, \forall l \in L_k, \forall k \in K \quad (10)$$

### C. The Saturation-Flow and Free-Flow Green-Band Constraints

Unlike most traditional approaches, which treat green-bands as outcomes based solely on signal parameters at adjacent intersections, the green-bands in this study are naturally derived from an analysis of vehicle movement from the origin to the destination.

The proposed model accounts for heterogeneous traffic conditions by differentiating local green bands into saturation-flow and free-flow bands (Fig. 7), reflecting traffic density variations. During phase initiation, vehicles discharge at saturation flow when queues are present and transition to free-flow conditions once queues dissipate. This approach enables more accurate queue length estimation by capturing the distinct queue formation characteristics associated with varying traffic densities.

For both green-bands, the widths are then decided by corresponding binary variables as indicated in (11)-(12) for outbound and (13)-(14) for inbound.

$$0 \leq b_{l,l',k}^s \leq \alpha_{l,l',k}^s C, \forall l \in L_k^o, \forall l' \in L_{k+1}^d, \forall k \in K^t \quad (11)$$

$$0 \leq b_{l,l',k}^f \leq \alpha_{l,l',k}^f C, \forall l \in L_k^o, \forall l' \in L_{k+1}^d, \forall k \in K^t \quad (12)$$

$$0 \leq \bar{b}_{l,l',k}^s \leq \bar{\alpha}_{l,l',k}^s C, \forall l \in \bar{L}_k^o, \forall l' \in \bar{L}_{k+1}^d, \forall k \in K^t \quad (13)$$

$$0 \leq \overline{b_{l,l',k}^f} \leq \overline{\alpha_{l,l',k}^f} C, \forall l \in \overline{L_k^o}, \forall l' \in \overline{L_{k+1}^d}, \forall k \in K^t \quad (14)$$

where  $\alpha_{l,l',k}^s(\overline{\alpha_{l,l',k}^s})$  is the binary variable indicating the existence of local outbound (inbound) saturation-flow green-band and the variable is equal to 1 if the green-band is greater than 0;  $\alpha_{l,l',k}^f(\overline{\alpha_{l,l',k}^f})$  is the binary variable indicating the existence of local outbound (inbound) free-flow green-band;  $b_{l,l',k}^s(\overline{b_{l,l',k}^s})$  is the variable indicating the bandwidth of the local outbound (inbound) saturation-flow green-band(s);  $b_{l,l',k}^f(\overline{b_{l,l',k}^f})$  is the variable indicating the bandwidth of the local outbound (inbound) free-flow green-band(s).

In order to calculate the widths of saturation-flow and free-flow green-bands, reference points are introduced according to Fig. 7. Constraint (15) presents  $o_{l,l',k}^1$ , which, in an analogical way, can be seen as the arrival time at the destination phase of the first vehicle from the queue at the origin phase.

$$o_{l,l',k}^1 = \theta_k + r_{l,k} + T_{l,l',k} + n_{l,l',k}C + T_{loss}, \quad \forall l \in \overline{L_k^o}, \forall l' \in \overline{L_{k+1}^d}, \forall k \in K^t \quad (15)$$

where  $\theta_k$  is the offset variable (s);  $r_{l,k}$  is the variable indicating the length of red signal before a phase(s);  $T_{l,l',k}(\overline{T_{l,l',k}})$  is the travel time of outbound (inbound) green-band(s);  $n_{l,l',k}(\overline{n_{l,l',k}})$  is the integer variable for the local outbound (inbound) green-band.

Similarly, constraints (16)-(17) are formulated to build other borders of arrival from origin phases. It is notable that the influence of over-saturation is also considered in (16). As the necessary discharging time is greater than the phase length for over-saturated phases,  $o_{l,l',k}^2$  and  $o_{l,l',k}^3$  refer to the identical point in the case, indicating that the bandwidth of free-flow will be reduced to 0 since all vehicles are discharged in saturation flow. Such idea is also adopted for the calculation of bandwidth to be presented next.

$$o_{l,l',k}^2 = o_{l,l',k}^1 + \min(u_{l,k}, \phi_{l,k}), \forall l \in \overline{L_k^o}, \forall l' \in \overline{L_{k+1}^d}, \forall k \in K^t \quad (16)$$

$$o_{l,l',k}^3 = o_{l,l',k}^1 + \phi_{l,k}, \forall l \in \overline{L_k^o}, \forall l' \in \overline{L_{k+1}^d}, \forall k \in K^t \quad (17)$$

Definitions for inbound reference points are also given as (18)-(20).

$$\overline{o_{l,l',k}^1} = \theta_{k+1} + r_{l',k+1} + \overline{T_{l,l',k}} + \overline{n_{l,l',k+1}}C + T_{loss}, \forall l \in \overline{L_k^o}, \forall l' \in \overline{L_{k+1}^d}, \forall k \in K^t \quad (18)$$

$$\overline{o_{l,l',k}^2} = \overline{o_{l,l',k}^1} + \min(u_{l',k+1}, \phi_{l',k+1}), \forall l \in \overline{L_k^o}, \forall l' \in \overline{L_{k+1}^d}, \forall k \in K^t \quad (19)$$

$$\overline{o_{l,l',k}^3} = \overline{o_{l,l',k}^1} + \phi_{l',k+1}, \forall l \in \overline{L_k^o}, \forall l' \in \overline{L_{k+1}^d}, \forall k \in K^t \quad (20)$$

Above definitions are built from the angle of origin phases. For destination phases, reference points are shown in (21)-(24) for both directions.

$$d_{l,l',k}^1 = \theta_{k+1} + r_{l',k+1} + u_{l',k+1} + n_{l,l',k+1}C, \forall l \in \overline{L_k^o}, \forall l' \in \overline{L_{k+1}^d}, \forall k \in K^t \quad (21)$$

$$d_{l,l',k}^2 = \theta_{k+1} + r_{l',k+1} + \phi_{l',k+1} + n_{l,l',k+1}C, \forall l \in \overline{L_k^o}, \forall l' \in \overline{L_{k+1}^d}, \forall k \in K^t \quad (22)$$

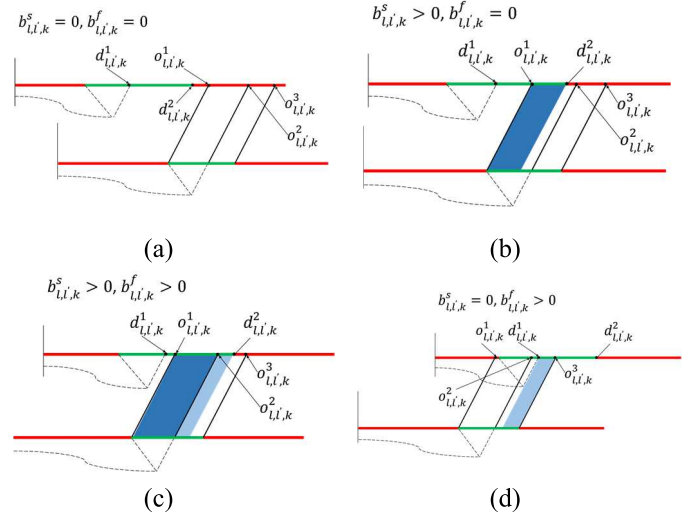


Fig. 8. Examples for local green-bands calculation using reference points.

$$\overline{d_{l,l',k}^1} = \theta_k + r_{l,k} + u_{l,k} + \overline{n_{l,l',k}}C, \forall l \in \overline{L_k^o}, \forall l' \in \overline{L_{k+1}^d}, \forall k \in K^t \quad (23)$$

$$\overline{d_{l,l',k}^2} = \theta_k + r_{l,k} + \phi_{l,k} + \overline{n_{l,l',k}}C, \quad \forall l \in \overline{L_k^o}, \forall l' \in \overline{L_{k+1}^d}, \forall k \in K^t \quad (24)$$

With the help of reference points, the borders of saturation-flow green-bands under different signal plans can be obtained. Some intuitive examples showing how reference points work are given in Fig. 8, according to which constraints (25)-(26) can be formulated, which defines the left and right borders of the saturation-flow green-band using reference points.

$$b_{l,l',k}^{s,f} = \max(o_{l,l',k}^1, d_{l,l',k}^1), \forall l \in \overline{L_k^o}, \forall l' \in \overline{L_{k+1}^d}, \forall k \in K^t \quad (25)$$

$$b_{l,l',k}^{s,b} = \min(o_{l,l',k}^2, d_{l,l',k}^2), \forall l \in \overline{L_k^o}, \forall l' \in \overline{L_{k+1}^d}, \forall k \in K^t \quad (26)$$

where  $b_{l,l',k}^{s,f}(\overline{b_{l,l',k}^{s,f}})$  is the variable indicating the left border of local outbound (inbound) saturation-flow green-band(s);  $b_{l,l',k}^{s,b}(\overline{b_{l,l',k}^{s,b}})$  is the variable indicating the right border of local outbound (inbound) saturation-flow green-band(s).

Constraint (27) shows that the bandwidth of saturation-flow green-band is the difference between the right and the left borders. Such constraints suffice to make allowance for both under- and over-saturation circumstances. When a destination phase is over-saturated, corresponding  $d_{l,l',k}^1$  and  $d_{l,l',k}^2$  will refer to the same point and thus there is no room for green-bands greater than 0.

$$\left| b_{l,l',k}^{s,f} - b_{l,l',k}^{s,b} + b_{l,l',k}^{s,f} \right| \leq (1 - \alpha_{l,l',k}^s)M, \forall l \in \overline{L_k^o}, \forall l' \in \overline{L_{k+1}^d}, \forall k \in K^t \quad (27)$$

Similar constraints for inbound can be presented as (28)-(30).

$$\overline{b_{l,l',k}^{s,f}} = \max(\overline{o_{l,l',k}^1}, \overline{d_{l,l',k}^1}), \forall l \in \overline{L_k^o}, \forall l' \in \overline{L_{k+1}^d}, \forall k \in K^t \quad (28)$$

$$\overline{b_{l,l',k}^{s,b}} = \min(\overline{o_{l,l',k}^2}, \overline{d_{l,l',k}^2}), \forall l \in \overline{L_k^o}, \forall l' \in \overline{L_{k+1}^d}, \forall k \in K^t \quad (29)$$

$$\left| \overline{b_{l,l',k}^{s,f}} - \overline{b_{l,l',k}^{s,b}} + \overline{b_{l,l',k}^{s,f}} \right| \leq (1 - \overline{\alpha_{l,l',k}^s})M, \forall l \in \overline{L_k^o}, \forall l' \in \overline{L_{k+1}^d}, \forall k \in K^t \quad (30)$$

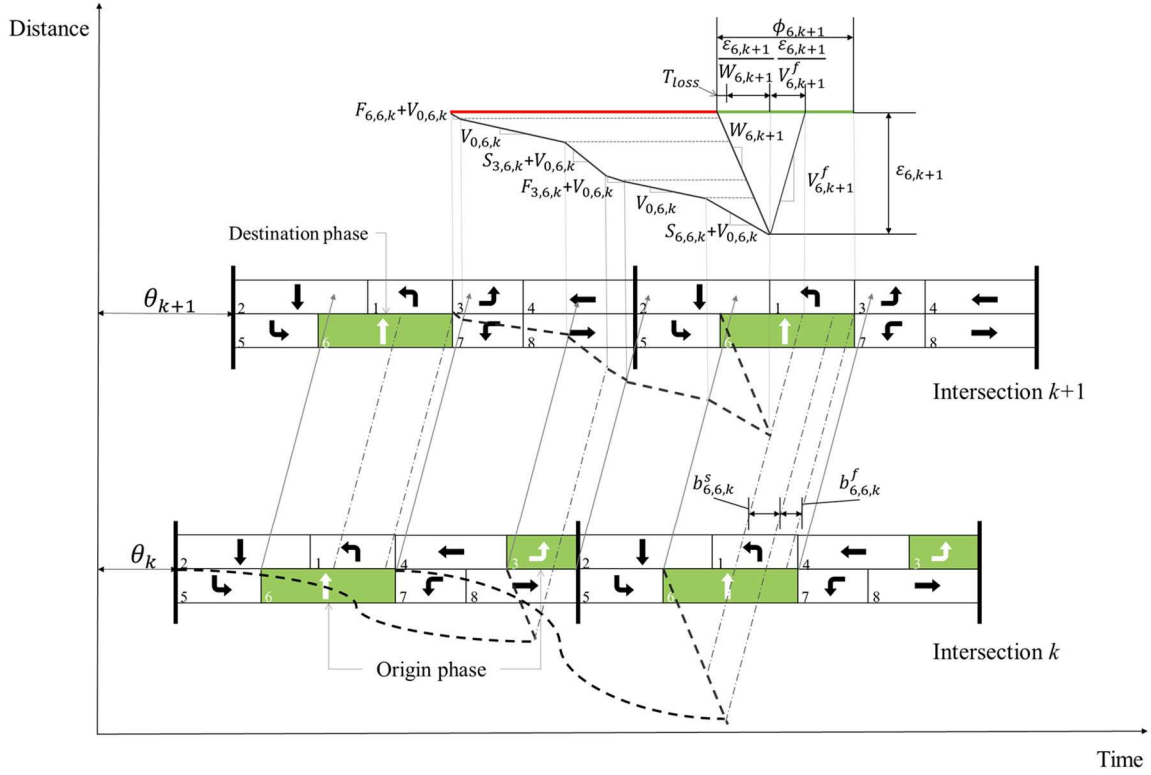


Fig. 9. The estimation on queue profile for a phase.

The discussion on the width of free-flow green-bands will not be extended and related constraints are built as (31)-(36) based on the same idea, using reference points  $o_{l,l',k}^2(o_{l,l',k}^2)$ ,  $o_{l,l',k}^3(o_{l,l',k}^3)$ ,  $d_{l,l',k}^1(d_{l,l',k}^1)$ , and  $d_{l,l',k}^2(d_{l,l',k}^2)$ .

$$b_{l,l',k}^{f,f} = \max(o_{l,l',k}^2, d_{l,l',k}^1), \forall l \in L_k^o, \forall l' \in L_{k+1}^d, \forall k \in K^t \quad (31)$$

$$b_{l,l',k}^{f,b} = \min(o_{l,l',k}^3, d_{l,l',k}^2), \forall l \in L_k^o, \forall l' \in L_{k+1}^d, \forall k \in K^t \quad (32)$$

$$\begin{aligned} |b_{l,l',k}^{f,f} - b_{l,l',k}^{f,b} + b_{l,l',k}^{f,f}| &\leq (1 - \alpha_{l,l',k}^f)M, \\ \forall l \in L_k^o, \forall l' \in L_{k+1}^d, \forall k \in K^t \end{aligned} \quad (33)$$

$$\overline{b_{l,l',k}^{f,f}} = \max(\overline{o_{l,l',k}^2}, \overline{d_{l,l',k}^1}), \forall l \in \overline{L_k^o}, \forall l' \in \overline{L_{k+1}^d}, \forall k \in K^t \quad (34)$$

$$\overline{b_{l,l',k}^{f,b}} = \min(\overline{o_{l,l',k}^3}, \overline{d_{l,l',k}^2}), \forall l \in \overline{L_k^o}, \forall l' \in \overline{L_{k+1}^d}, \forall k \in K^t \quad (35)$$

$$\begin{aligned} |\overline{b_{l,l',k}^{f,f}} - \overline{b_{l,l',k}^{f,b}} + \overline{b_{l,l',k}^{f,f}}| &\leq (1 - \overline{\alpha_{l,l',k}^f})M, \\ \forall l \in \overline{L_k^o}, \forall l' \in \overline{L_{k+1}^d}, \forall k \in K^t \end{aligned} \quad (36)$$

where  $b_{l,l',k}^{f,f}(b_{l,l',k}^{f,f})$  is the variable indicating the left border of local outbound (inbound) free-flow green-band(s);  $b_{l,l',k}^{f,b}(\overline{b_{l,l',k}^{f,b}})$  is the variable indicating the right border of local outbound (inbound) free-flow green-band(s).

#### D. The Stop Estimation Constraints

Unlike conventional band-based models that assume pre-determined queue discharging times, the proposed model accounts for the inter-correlation of queue profiles across phases at adjacent intersections. Specifically, changes in upstream queues—potentially resulting from altered signal

plans—are considered to affect the profile of downstream queues. Additionally, it is assumed that vehicles only stop when they are compelled to join the queue at under-saturated phases. Consequently, the maximum queue length for under-saturated phases within a given cycle is treated as equivalent to the number of vehicular stops in the same cycle.

The estimation on the queue profile for a phase is illustrated by an example given in Fig. 9, where the queue for phase 6 at intersection  $k+1$  is comprised of traffic flow from three origins, upstream through, upstream left-turning and upstream right-turning, namely. As aforementioned in the assumptions, for upstream through and left-turning, the traffic arrival is assumed to be approximately uniform for both local saturation flow and free flow. For upstream right-turning, the traffic arrival is thought to be uniform within the whole cycle. The number of vehicles from an origin which would join the queue and be forced to stop can be calculated according to the corresponding local saturation-flow and free-flow green-bands.

According to the example in Fig. 9 the composition of the max queue length for a phase is derived as constraint (37), suggesting that the maximum queue length consists of vehicles from three origins, which are right-turning, left-turning and through phases at upstream intersections.

$$\varepsilon_{l',k+1} = \sum_l q_{l,l',k}, \forall l \in L_k^o \cup L_k^r, \forall l' \in L_{k+1}^d, \forall k \in K^t \quad (37)$$

where  $q_{l,l',k}$  is the variable indicating the outbound queue length contributed by phase  $l$  at intersection  $k$  for phase  $l'$  at intersection  $k+1$  (m).

Under the presumption that vehicles would be forced to join the queue unless they move within local green-bands, the queue length caused by through or left-turning vehicles is represented by constraint (38) when origin phases are under-saturated. Specifically, it is also denoted in (38) that the queue length from a single origin can be further divided into two parts, where one is contributed by saturation flow and the other attributed to free flow.

$$\begin{aligned} & \left| q_{l,l',k} - S_{l,l',k}(u_{l,k} - b_{l,l',k}^s) - F_{l,l',k}(\phi_{l,k} - u_{l,k} - b_{l,l',k}^f) \right| \\ & \leq \psi_{l,k}M, \\ & \forall l \in L_k^o, \forall l' \in L_{k+1}^d, \forall k \in K^t \end{aligned} \quad (38)$$

where  $S_{l,l',k}(\overline{S_{l,l',k}})$  is the outbound (inbound) queue accumulating speed for saturation flow(m/s);  $F_{l,l',k}(\overline{F_{l,l',k}})$  is the outbound (inbound) queue accumulating speed for free flow(m/s).

Still, the queue length demands different presentations for origin phases facing over-saturation, where queue clearing time exceeds the phase length. As such, following constraint (39) are provided, which shows that the queue contributed by over-saturated origin phases consists of only saturation flow.

$$\begin{aligned} & \left| q_{l,l',k} - S_{l,l',k}(\phi_{l,k} - b_{l,l',k}^s) \right| \leq (1 - \psi_{l,k})M, \forall l \in L_k^o, \\ & \forall l' \in L_{k+1}^d, \forall k \in K^t \end{aligned} \quad (39)$$

As for uncontrolled right-turning movements, whether they join the queue or not is decided by the saturation level of the destination phases. When under-saturated, all right-turning vehicles which travel through the destination phase after the queue is fully discharged will not be stopped as suggested in constraint (40).

$$\begin{aligned} & \left| q_{l,l',k} - F_{l,l',k}(C - \phi_{l',k+1} + u_{l',k+1}) \right| \leq \psi_{l',k+1}M, \forall l \\ & \in L_k^t, \forall l' \in L_{k+1}^d, \forall k \in K^t \end{aligned} \quad (40)$$

Otherwise, over-saturated destination phases will force all right-turning vehicles to stop (constraint (41)).

$$\begin{aligned} & \left| q_{l,l',k} - F_{l,l',k}C \right| \leq (1 - \psi_{l',k+1})M, \forall l \in L_k^t, \forall l' \in L_{k+1}^d, \\ & \forall k \in K^t \end{aligned} \quad (41)$$

On the basis of similar discussions, constraints (42)-(46) are established for inbound.

$$\varepsilon_{l,k} = \sum_{l'} \overline{q_{l,l',k}}, \forall l \in \overline{L_k^o}, \forall l' \in \overline{L_{k+1}^d} \cup \overline{L_{k+1}^r}, \forall k \in K^t \quad (42)$$

$$\begin{aligned} & \left| \overline{q_{l,l',k}} - \overline{S_{l,l',k}}(u_{l',k+1} - b_{l,l',k}^s) - \overline{F_{l,l',k}}(\phi_{l',k+1} - u_{l',k+1} - b_{l,l',k}^f) \right| \\ & \leq \psi_{l',k+1}M, \forall l \in \overline{L_k^o}, \forall l' \in \overline{L_{k+1}^d}, \forall k \in K^t \end{aligned} \quad (43)$$

$$\begin{aligned} & \left| \overline{q_{l,l',k}} - \overline{S_{l,l',k}}(\phi_{l',k+1} - b_{l,l',k}^s) \right| \leq (1 - \psi_{l',k+1})M, \forall l \in \overline{L_k^o}, \\ & \forall l' \in \overline{L_{k+1}^d}, \forall k \in K^t \end{aligned} \quad (44)$$

$$\begin{aligned} & \left| \overline{q_{l,l',k}} - \overline{F_{l,l',k}}(C - \phi_{l,k} + u_{l,k}) \right| \leq \psi_{l,k}M, \forall l \in \overline{L_k^o}, \\ & \forall l' \in \overline{L_{k+1}^r}, \forall k \in K^t \end{aligned} \quad (45)$$

$$\begin{aligned} & \left| \overline{q_{l,l',k}} - \overline{F_{l,l',k}}C \right| \leq (1 - \psi_{l,k})M, \forall l \in \overline{L_k^o}, \forall l' \in \overline{L_{k+1}^r}, \\ & \forall k \in K^t \end{aligned} \quad (46)$$

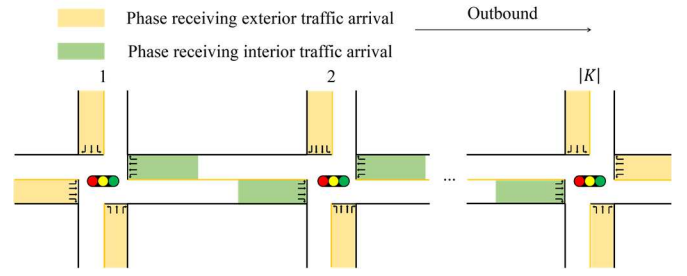


Fig. 10. Illustrations for (a) phases receiving exterior and interior traffic arrival.

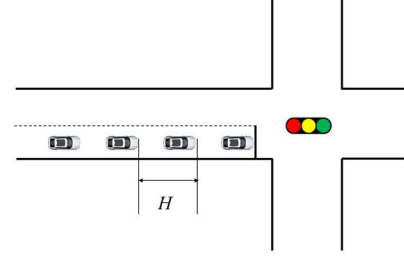


Fig. 11. Illustrations for space headway.

where  $\overline{q_{l,l',k}}$  is the variable indicating the inbound queue length contributed by phase  $l'$  at intersection  $k+1$  for phase  $l$  at intersection  $k$ (m).

Aside from the formulations presented above, it is also necessary to make allowance for the cases regarding the queue profile estimation for phases where all arriving vehicles come from outside the artery, as shown in Fig. 10.

Specifically, these vehicles are assumed to arrive uniformly over time as their origins are out of reach. As such, constraints are built as (47)-(49) for under-saturated destination phases. It is notable that vehicle space headway (Fig. 11) is included in constraints (47)-(49) in order to build the connection between traffic flow and the maximum queue length.

$$\left| \varepsilon_{l,1} - V_{l,1}H(C - \phi_{l,1} + u_{l,1}) \right| \leq \psi_{l,1}M, \forall l \in L_1^d \quad (47)$$

$$\left| \varepsilon_{l,|K|} - V_{l,|K|}H(C - \phi_{l,|K|} + u_{l,|K|}) \right| \leq \psi_{l,|K|}M, \forall l \in \overline{L_{|K|}^o} \quad (48)$$

$$\left| \varepsilon_{l,k} - V_{l,k}H(C - \phi_{l,k} + u_{l,k}) \right| \leq \psi_{l,k}M, \forall l \in L_u, \forall k \in K \quad (49)$$

where  $V_{l,k}$  is the average flow for a phase(pcu/s);  $H$  is the average space headway of queue(m).

For over-saturated phases, constraints (50)-(52) will be effective.

$$\left| \varepsilon_{l,1} - V_{l,1}HC \right| \leq (1 - \psi_{l,1})M, \forall l \in L_1^d \quad (50)$$

$$\left| \varepsilon_{l,|K|} - V_{l,|K|}HC \right| \leq (1 - \psi_{l,|K|})M, \forall l \in \overline{L_{|K|}^o} \quad (51)$$

$$\left| \varepsilon_{l,k} - V_{l,k}HC \right| \leq (1 - \psi_{l,k})M, \forall l \in L_u, \forall k \in K \quad (52)$$

Constraints (53)-(54) indicate the speed of queue accumulation by saturation flow and free flow for outbound ( $S_{l,l',k}$  and  $F_{l,l',k}$ ), respectively. As vehicles would go to different destination phases, the queue accumulating speed for saturation flow from an origin is calculated according to constraint (53), where trajectories between phase pairs are used as weights.

$$S_{l,l',k} = \frac{Y_{l,l',k}}{N_{l,k}} W_{l,k}, \forall l \in L_k^o, \forall l' \in L_{k+1}^d, \forall k \in K^t \quad (53)$$

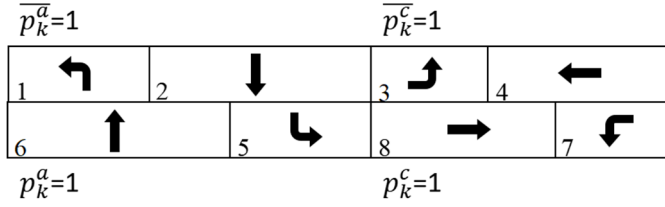


Fig. 12. An example for phase sequence.

$$F_{l,l',k} = \frac{Y_{l,l',k}}{N_{l,k}} m_{l,k}, \forall l \in L_k^o, \forall l' \in L_{k+1}^d, \forall k \in K' \quad (54)$$

where  $Y_{l,l',k}(\overline{Y_{l,l',k}})$  is the number of collected outbound (inbound) trajectories;  $N_{l,k}$  is the number of collected trajectories;  $m_{l,k}$  is the free flow variable (m/s);  $K'$  is the subset of  $K$ , where the last intersection is excluded.

Similarly, constraints (55) and (56) are built for inbound notations.

$$\overline{S_{l,l',k}} = \frac{\overline{Y_{l,l',k}}}{N_{l',k+1}} W_{l',k+1}, \forall l \in \overline{L_k^o}, \forall l' \in \overline{L_{k+1}^d}, \forall k \in K' \quad (55)$$

$$\overline{F_{l,l',k}} = \frac{\overline{Y_{l,l',k}}}{N_{l',k+1}} m_{l',k+1}, \forall l \in \overline{L_k^o}, \forall l' \in \overline{L_{k+1}^d}, \forall k \in K' \quad (56)$$

Unlike  $W_{l,k}$ , the free flow released from an origin phase ( $m_{l,k}$ ) is regarded as a variable. The intention is given as follows: the maximum queue length for a phase will vary under different signal plans, resulting in the change of the number of vehicles which leave the intersection in saturation flow. Still, the number of total vehicles passing through a phase would not change if no over-saturation is observed at the origin phase. Thus, the free flow for a phase would be forced to vary, as displayed in constraint (57). While it is also quite plain that the free flow of an over-saturated origin phase is equal to 0, as shown in (58). In the meantime, for right-turning movement, (59) is given which estimates  $m_{l,k}$  as a constant.

$$|W_{l,k}(u_{l,k} - T_{loss}) + m_{l,k}(\phi_{l,k} - u_{l,k}) - V_{l,k}H| \leq y_{l,k}M, \quad \forall l \in L_k^o \cup \overline{L_k^d}, \forall k \in K \quad (57)$$

$$|m_{l,k}| \leq (1 - y_{l,k})M, \forall l \in L_k^o \cup \overline{L_k^d}, \forall k \in K \quad (58)$$

$$m_{l,k} = V_{l,k}H, \forall l \in L_k^i \cup \overline{L_k^r}, \forall k \in K \quad (59)$$

However, it is important to clarify that the maximum queue length defined above refers to the maximum incremental queue length that can accumulate within a cycle. For over-saturated phases, this queue length is not stable from cycle to cycle and will progressively increase over time. In these cases, the queue length is not directly equivalent to the total number of vehicle stops. Nonetheless, this approach still contributes to improving system efficiency and safety. When a phase is over-saturated, the focus of optimization shifts toward reducing the level of saturation, thus mitigating potential inefficiencies and enhancing overall system performance.

### E. The Phase Sequence Constraints

The strict definition of the binary variables for phase sequence is provided as follows and Fig. 12 serves as a more

intuitive example of phase sequence.

$$p_k^j = \begin{cases} 1, & \text{if outbound through is} \\ & \text{before inbound left} \end{cases}, j \in \{a, c\}, \forall k \in K$$

$$\overline{p_k^j} = \begin{cases} 1, & \text{if outbound left is} \\ & \text{before inbound through} \end{cases}, j \in \{a, c\}, \forall k \in K$$

$$0, \text{o.w.}$$

The phase sequence constraints are exhibited according to Fig. 12. The constraints for phase sequence are built in such way so as not to hurt the linearity of the proposed model. These constraints enable the phase sequence to change and define how the red signal between origin and destination phases would vary as the phase sequence varies. Specifically, looking at the first ring, constraints (60)-(61) are for phase 1, (62)-(63) for phase 2, and (64)-(65) for phase 3.

$$|r_{1,k}| \leq (1 - \overline{p_k^a})M, \forall k \in K \quad (60)$$

$$|r_{1,k} - \phi_{2,k}| \leq \overline{p_k^a}M, \forall k \in K \quad (61)$$

$$|r_{2,k}| \leq \overline{p_k^a}M, \forall k \in K \quad (62)$$

$$|r_{2,k} - \phi_{1,k}| \leq (1 - \overline{p_k^a})M, \forall k \in K \quad (63)$$

$$|r_{3,k} - \phi_{1,k} - \phi_{2,k}| \leq (1 - \overline{p_k^c})M, \forall k \in K \quad (64)$$

$$|r_{3,k} - \phi_{1,k} - \phi_{2,k} - \phi_{4,k}| \leq \overline{p_k^c}M, \forall k \in K \quad (65)$$

Constraints (66)-(71) are similarly formulated for phases on the other ring.

$$|r_{5,k} - \phi_{6,k}| \leq (1 - p_k^a)M, \forall k \in K \quad (66)$$

$$|r_{5,k}| \leq p_k^aM, k \in K, \forall k \in K \quad (67)$$

$$|r_{6,k}| \leq (1 - p_k^a)M, \forall k \in K \quad (68)$$

$$|r_{6,k} - \phi_{5,k}| \leq p_k^aM, \forall k \in K \quad (69)$$

$$|r_{7,k} - \phi_{5,k} - \phi_{6,k}| \leq p_k^cM, \forall k \in K \quad (70)$$

$$|r_{7,k} - \phi_{5,k} - \phi_{6,k} - \phi_{8,k}| \leq (1 - p_k^c)M, \forall k \in K \quad (71)$$

### F. Solution Algorithm

The proposed model is in fact a MINLP because of constraints (16), (19), (25)-(26), (28)-(29), (31)-(32), (34)-(35), (38) and (57). These non-linear constraints can be further categorized into two types: (16), (19), (25)-(26), (28)-(29), (31)-(32), and (34)-(35) are min/max constraints; (38), (43), and (57) are bilinear constraints.

1) *The Linearization of Max/Min Constraints:* For min/max constraints, they can be linearized by using extra binary variables. To save room, (25) is taken as an example, which can be transformed into linear constraints (72)-(76).

$$b_{l,l',k}^{s,f} \leq o_{l,l',k}^1, \forall l \in L_k^o, \forall l' \in L_{k+1}^d, \forall k \in K' \quad (72)$$

$$b_{l,l',k}^{s,f} \leq d_{l,l',k}^1, \forall l \in L_k^o, \forall l' \in L_{k+1}^d, \forall k \in K' \quad (73)$$

$$b_{l,l',k}^{s,f} \geq o_{l,l',k}^1 - (1 - x_{l,l',k}^o)M, \forall l \in L_k^o, \forall l' \in L_{k+1}^d, \forall k \in K' \quad (74)$$

$$b_{l,l',k}^{s,f} \geq d_{l,l',k}^1 - (1 - x_{l,l',k}^d)M, \forall l \in L_k^o, \forall l' \in L_{k+1}^d, \forall k \in K' \quad (75)$$

$$x_{l,l',k}^o + x_{l,l',k}^d \geq 1, \forall l \in L_k^o, \forall l' \in L_{k+1}^d, \forall k \in K' \quad (76)$$

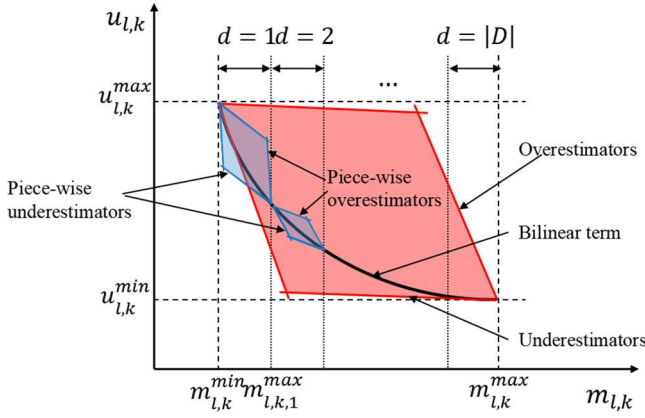


Fig. 13. Illustrations for piece-wise McCormick envelope.

where  $x_{l,l',k}^o$  is the relaxing binary variable for  $o_{l,l',k}^1$ ;  $x_{l,l',k}^d$  is the relaxing binary variable for  $d_{l,l',k}^1$ .

2) *Piece-Wise McCormick Envelopes*: McCormick envelopes (McCormick 1976) are employed in the study to provide linear substitutes for bilinear constraints. Intermediate variables are first declared for bilinear constraints as (77)-(80).

$$\zeta_{l,k} = m_{l,k} \phi_{l,k}, \forall l \in L_k^o \cup \bar{L}_k^d, \forall k \in K \quad (77)$$

$$z_{l,k} = m_{l,k} u_{l,k}, \forall l \in L_k^o \cup \bar{L}_k^d, \forall k \in K \quad (78)$$

$$\tau_{l,l',k} = m_{l,k} b_{l,l',k}^f, \forall l \in L_k^o, \forall l' \in L_{k+1}^d, \forall k \in K \quad (79)$$

$$\bar{\tau}_{l,l',k} = m_{l',k+1} b_{l,l',k}^f, \forall l \in \bar{L}_k^o, \forall l' \in \bar{L}_{k+1}^d, \forall k \in K \quad (80)$$

Constraints (38) and (57) are then transformed to linear constraints (81)-(83).

$$W_{l,k}(u_{l,k} - T_{loss}) + \zeta_{l,k} - V_{l,k} H_{l,k} = z_{l,k}, \forall l \in L_k^o \cup \bar{L}_k^d, \forall k \in K \quad (81)$$

$$\left| q_{l,l',k} - S_{l,l',k}(u_{l,k} - b_{l,l',k}^s) - \frac{Y_{l,l',k}}{N_{l,k}}(\zeta_{l,k} - z_{l,k} - \tau_{l,l',k}) \right| \leq y_{l,k} M, \forall l \in L_k^o, \forall l' \in L_{k+1}^d, \forall k \in K^t \quad (82)$$

$$\left| \bar{q}_{l,l',k} - \bar{S}_{l,l',k}(u_{l',k+1} - \bar{b}_{l,l',k}^s) - \frac{\bar{Y}_{l,l',k}}{N_{l',k+1}}(\zeta_{l',k+1} - z_{l',k+1} - \bar{\tau}_{l,l',k}) \right| \leq y_{l',k+1} M, \forall l \in \bar{L}_k^o, \forall l' \in \bar{L}_{k+1}^d, \forall k \in K^t \quad (83)$$

In fact, the feasible region of the adjusted constraints may not be tight enough and the solution may not be feasible because of possible violation of the bilinear constraints. Thus, a piece-wise McCormick envelope is adopted to further tighten the feasible regions. The main idea of piece-wise McCormick envelope is to partition the domain of a variable in the bilinear constraints into several disjoint regions, among which an optimal region is selected, and, therefore, a tightened region is obtained. A simplified illustration is presented in Fig. 13.

The uniform partition [33] is employed here whereas other methods, such as adaptive segment length [34], are also applicable. Before partition is administered, it should be decided first which variable would be chosen for the uniform partition. In the case of the study, probable choices include  $m_{l,k}$ ,  $u_{l,k}$  and the combination of both variables. As practice suggests,

partition should only be applied to one dimension of the bilinear terms in order that the demand for computational capability remains acceptable. However, the search for the best choice is beyond the scope of the study and further discussions will not be extended. To save room, only (78) is taken as an example for the McCormick envelope, as shown in constraints (84)-(87), where  $m_{l,k}$  is selected for further division.

$$z_{l,k} \geq \sum_{d=1}^{|D|} (m_{l,k,d}^{\min} u_{l,k,d} + m_{l,k,d} u_{l,k}^{\min} - m_{l,k,d}^{\min} u_{l,k}^{\min} \lambda_{l,k,d}), \quad (84)$$

$$\forall l \in L_k^o \cup \bar{L}_k^d, \forall k \in K$$

$$z_{l,k} \geq \sum_{d=1}^{|D|} (m_{l,k,d}^{\max} u_{l,k,d} + m_{l,k,d} u_{l,k}^{\max} - m_{l,k,d}^{\max} u_{l,k}^{\max} \lambda_{l,k,d}), \quad (85)$$

$$\forall l \in L_k^o \cup \bar{L}_k^d, \forall k \in K$$

$$z_{l,k} \leq \sum_{d=1}^{|D|} (m_{l,k,d}^{\max} u_{l,k,d} + m_{l,k,d} u_{l,k}^{\min} - m_{l,k,d}^{\max} u_{l,k}^{\min} \lambda_{l,k,d}), \quad (86)$$

$$\forall l \in L_k^o \cup \bar{L}_k^d, \forall k \in K$$

$$z_{l,k} \leq \sum_{d=1}^{|D|} (m_{l,k,d}^{\min} u_{l,k,d} + m_{l,k,d} u_{l,k}^{\max} - m_{l,k,d}^{\min} u_{l,k}^{\max} \lambda_{l,k,d}), \quad (87)$$

$$\forall l \in L_k^o \cup \bar{L}_k^d, \forall k \in K$$

where  $m_{l,k,d}$  is the variable indicating the value of  $m_{l,k}$  in segment  $d$ ;  $m_{l,k,d}^{\min}$  ( $m_{l,k,d}^{\max}$ ) is the minimum (maximum) of  $m_{l,k,d}$ ;  $\lambda_{l,k,d}$  is the binary variable indicating whether  $m_{l,k,d}$  falls within segment  $d$ ,  $\lambda_{l,k,d}$  is equal to 1 if  $m_{l,k,d}$  falls within segment  $d$ ;  $u_{l,k,d}$  is the variable indicating the value of  $u_{l,k}$  in segment  $d$ ;  $D$  is the set of segments and  $|D|$  is the cardinality of the set.

To partition the value of involved variables, more constraints are given as (88)-(94). Equations (88) and (89) show how the specific values of the variables are decided. Constraint (90) indicates that only one segment can be chosen during the solving process. The upper and lower bounds of each segment for  $m_{l,k,d}$  are defined in (91) and (92), respectively. Constraints (93) and (94) suggest that only the bounds of the selected segment are effective.

$$m_{l,k} = \sum_{d=1}^{|D|} m_{l,k,d}, \forall l \in L_k^o \cup \bar{L}_k^d, \forall k \in K \quad (88)$$

$$u_{l,k} = \sum_{d=1}^{|D|} u_{l,k,d}, \forall l \in L_k^o \cup \bar{L}_k^d, \forall k \in K \quad (89)$$

$$\sum_{d=1}^{|D|} \lambda_{l,k,d} = 1, \forall l \in L_k^o \cup \bar{L}_k^d, \forall k \in K \quad (90)$$

$$m_{l,k,d}^{\min} = m_{l,k}^{\min} + \frac{(m_{l,k}^{\max} - m_{l,k}^{\min})(d-1)}{|D|}, \forall l \in L_k^o \cup \bar{L}_k^d, \forall k \in K, d \in D \quad (91)$$

$$m_{l,k,d}^{\max} = m_{l,k}^{\min} + \frac{(m_{l,k}^{\max} - m_{l,k}^{\min})d}{|D|}, \forall l \in L_k^o \cup \bar{L}_k^d, \forall k \in K, d \in D \quad (92)$$

$$m_{l,k,d}^{\min} \lambda_{l,k,d} \leq m_{l,k,d} \leq m_{l,k,d}^{\max} \lambda_{l,k,d}, \forall l \in L_k^o \cup \bar{L}_k^d, \forall k \in K, \forall d \in D \quad (93)$$

$$u_{l,k,d}^{\min} \lambda_{l,k,d} \leq u_{l,k,d} \leq u_{l,k,d}^{\max} \lambda_{l,k,d}, \forall l \in L_k^o \cup \bar{L}_k^d, \forall k \in K, \forall d \in D \quad (94)$$

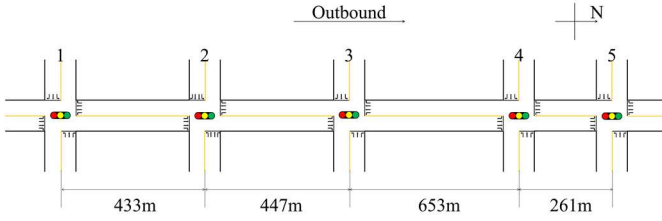


Fig. 14. The geometry of the road segment.

where  $u_{l,k,d}^{\max}$  ( $u_{l,k,d}^{\min}$ ) is the variable indicating the maximum(minimum) of  $u_{l,k}$  in segment  $d$

The use of a piecewise McCormick envelope provides a more compact method for exploring the solution space around the original feasible region defined by the bilinear constraints. However, it is important to note that increasing the number of partitions—thereby tightening the feasible regions—also results in a higher number of intermediate variables, which in turn escalates the computational burden. This trade-off between solution accuracy and computational efficiency must be carefully considered in the model implementation. Generally, the technique provides satisfactory performance with three partitions [33].

After using the above techniques, the proposed model is transformed into a MILP, which can be solved by many commercial solvers such as CPLEX and GUROBI.

#### IV. NUMERICAL EXAMPLES

##### A. Experiment Design

Numerical examples are given with a road segment located in the City of Rizhao, Shandong Province, China. As shown in Fig. 14, the segment is comprised of five intersections.

Specifically, Yang's model [24] and MP-BAND [32] are chosen for comparison. All models are accessible to the complete OD information of all scenarios so as to rule out the effect caused by their knowledge of the traffic demand. The solving of all models is carried out by GUROBI 10.0.

Key parameters adopted in the simulation are listed below:

- (1) The total simulation time is 3700s, with the first 50s and the last 50s excluded from the collection of MOEs (measures of effectiveness);
- (2) The simulation step is 1s, which is also the minimum sampling interval.
- (3) The time loss for each phase is 4s.
- (4) The average vehicle length is 3.5m.
- (5) The saturation flow rate for all approaches at all intersections is 1800 pcu/h.
- (6) The free flow speed is set as 16m/s.
- (7) The average space headway is 6m.
- (8) The minimum path bandwidth for Yang's model is set as 8s;
- (9) The minimum local bandwidth and connected bandwidth for MP-BAND are set as 5s and 2.5s for scenarios where uniform cycle length is no more than 85s. For scenarios with greater cycles, 8s and 4s are adopted.
- (10) Multiple rounds of simulations are necessary to account for the randomness in real-world traffic systems [35].

TABLE III  
SCENARIO INFORMATION

| Index | Description                                                                                            | Uniform cycle (s) |
|-------|--------------------------------------------------------------------------------------------------------|-------------------|
| A0    | Basic scenario with volume concentrated on multiple short paths.                                       | 70                |
| A1    | Scenario A0 * 1.10                                                                                     | 80                |
| A2    | Scenario A0 * 1.25                                                                                     | 85                |
| A3    | Scenario A0 * 1.50                                                                                     | 115               |
| A4    | Scenario A0 * 1.80 (with slightly over-saturated phase 3 at intersection 2)                            | 160               |
| B0    | Basic scenario with volume concentrated on multiple long paths                                         | 65                |
| B1    | Scenario B0 * 1.20                                                                                     | 85                |
| B2    | Scenario B0 * 1.40                                                                                     | 100               |
| B3    | Scenario B0 * 1.60                                                                                     | 105               |
| B4    | Scenario A0 * 1.80 (with slightly over-saturated phases 6 and 8 at intersection 2)                     | 150               |
| C0    | Basic scenario with two-way demand and heavy cross-street traffic at intersection 2 and intersection 4 | 80                |
| C1    | Scenario C0*1.10                                                                                       | 95                |
| C2    | Scenario C0*1.20                                                                                       | 110               |
| C3    | Scenario C0*1.35                                                                                       | 130               |
| C4    | Scenario C0*1.55(with slightly over-saturated phases 4 and 8 at intersections 2 and 4)                 | 160               |

Convergence analysis based on the scenarios was conducted, which demonstrated that the results stabilize after 80 replications per scenario. This number is further validated by a related study [36], ensuring the reliability and robustness of the findings.

To fully evaluate the efficacy of the proposed model, fifteen scenarios are built in SUMO. Detailed information of the scenarios is shown in TABLE III. The uniform cycles are decided as the maximum among the cycles for each intersection, which are produced beforehand by SYNCHRO, along with phase splits. Three scenarios with over-saturated phases are also introduced.

The baseline traffic demand scenarios (A0, B0, and C0), obtained from real-world detectors, are provided in the supplementary material. Scenarios A0 and B0 are concentrated primarily on arterial multiple paths, whereas scenario C0 incorporates typical two-way demand alongside substantial cross-street through traffic. Fig. 15 illustrates the target paths for Yang's model. Notably, paths 3-6 in Fig. 15(c) represent cross-street movements that, while not compatible with Yang's model due to its arterial-focused formulation, are included to put an emphasis on the demand in scenario group C.

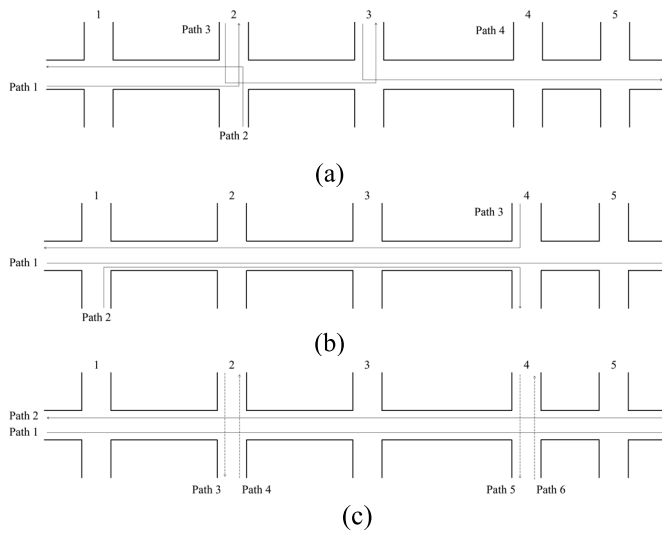


Fig. 15. Path nomination for Yang's model in (a) scenarios A (b) scenarios B (c) scenarios C.

TABLE IV  
SOLVING TIME COMPARISON

| Model description       | Gap (%) | Average Solving time(s) |
|-------------------------|---------|-------------------------|
| Yang's model            | 0.1     | 7.3                     |
| MP-BAND                 | 0.1     | 23.5                    |
| The proposed model      | 5       | 255.2                   |
| without McCormick       | 1       | 1873.8                  |
| envelope                | 0.1     | >9999                   |
| The proposed model      | 0.1     | 89.7                    |
| with McCormick envelope |         |                         |

### B. Solving Efficiency and Scalability

As the proposed model is formulated as a MINLP in the first place and then linearized with the help of piecewise McCormick envelopes, experiments regarding solving efficiency should be conducted.

All models shown in TABLE IV are solved by GUROBI 10.0 for ten times for all scenarios to gain an average solving time. The gap in the table is a parameter in GUROBI indicating the upper bound on the actual gap of the final solution. The parameter controls the minimal quality of the returned solution. As a matter of fact, small gaps usually result in better solutions but tend to make the solving process more time-consuming.

As indicated in TABLE IV, a reduction in the optimality gap leads to a substantial increase in computational time for the proposed model when the McCormick envelope is not applied. Specifically, when the gap is constrained to no more than 0.1%, the required solving time becomes prohibitively long, rendering the model nearly infeasible. In contrast, the solving times for other models—including the proposed model incorporating the McCormick envelope—remain within the same

TABLE V

THE NUMBER OF VARIABLES AND CONSTRAINTS IN THE PROPOSED MODEL WITH PIECE-WISE MCCORMICK ENVELOPES

|                      | Number         |
|----------------------|----------------|
| Continuous variables | $324 K  - 120$ |
| Integer variables    | $8 K  - 8$     |
| Binary variables     | $62 K  - 36$   |
| Linear constraints   | $451 K  - 312$ |

TABLE VI

THE SOLVING TIME OF THE PROPOSED MODEL WITH PIECE-WISE MCCORMICK ENVELOPES

| Number of intersections | Average solving time(s) |
|-------------------------|-------------------------|
| 7                       | 119.6                   |
| 8                       | 135.2                   |
| 9                       | 159.3                   |
| 10                      | 187.4                   |

order of magnitude, demonstrating computational efficiency suitable for fixed-time signal coordination control.

It should be noted that these computational times were obtained for a scenario involving five intersections, a configuration widely regarded as appropriate for signal coordination [37], [38]. Previous studies [37] suggest that partitioning strategies should be employed for arteries exceeding six intersections. Nevertheless, given that optimization may still be required for longer arteries, TABLE V provides an estimation of the number of variables and constraints in the proposed model when using piecewise McCormick envelopes with three partitions. For simplicity, all intersections were assumed to be four-armed and governed by the NEMA ring-barrier scheme.

As shown in TABLE V, the number of variables and constraints remains within the order of 104 for arteries comprising up to twenty intersections—a scale substantially larger than typical real-world scenarios. According to [39], commercial optimization solvers such as Gurobi and CPLEX can efficiently compute optimal solutions for problems with up to 10,000 variables and constraints, typically requiring only seconds to minutes.

Additional calculations were performed for scenarios involving more intersections, though the detailed results are omitted to conserve space. The solving times for these scenarios are presented in TABLE VI, which aligns with the conclusions drawn earlier. Given that fixed-time signal coordination is typically implemented over extended periods (ranging from tens of minutes to hours), the proposed model demonstrates sufficient computational tractability and scalability for practical applications.

### C. Solution Comparison

To save room, only the solutions produced by the proposed model and MP-BAND for scenarios A0 are illustrated in

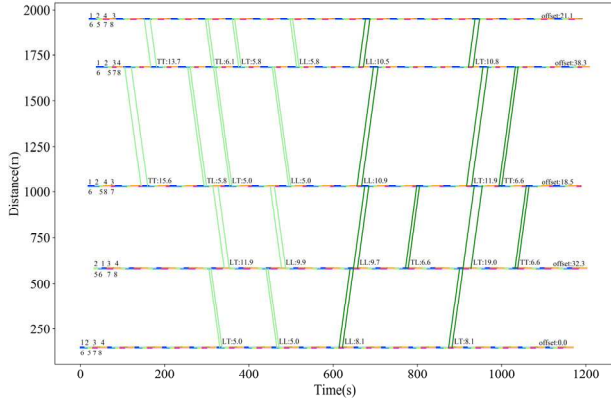


Fig. 16. Time-space diagram by MP-BAND for scenario A0.

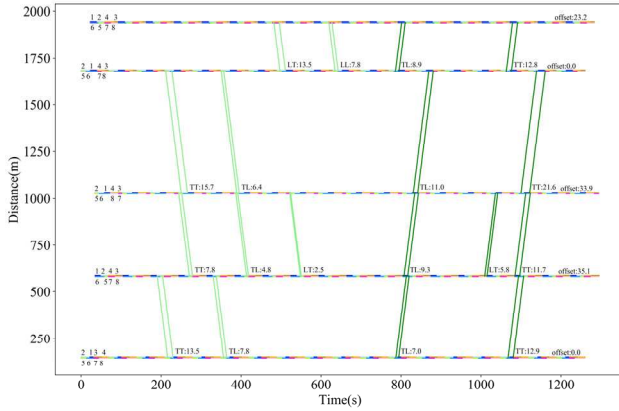


Fig. 17. Time-space diagram by the proposed model for scenario A0.

Fig. 16 and in Fig. 17 for comparison. MP-BAND is formulated with the objective of weighted sum of local bandwidths without considering the change of queue profile. Therefore, many hyperparameters have to be offered beforehand, such as the minimum bandwidth for local progression, resulting in the minimum bandwidth of 5s in Fig. 16. However, such predetermined minimum bandwidths may hinder the identification of more effective signal plans, as illustrated in Fig. 17, where the proposed model produces bandwidths smaller than the prescribed minimum.

It is important to note that, since queue discharge times are not explicitly considered, the local progression bandwidth in MP-BAND differs conceptually from that in the proposed model. Therefore, the bandwidths generated by our model should be interpreted as relatively smaller only in comparison to empirical minimum values reported in previous studies and engineering practices. Nevertheless, the ability of the proposed model to generate bandwidths that deviate from empirical expectations—alongside its explicit consideration of variable queue accumulation and discharge—is a central feature of its design. This reflects a system-optimal perspective, wherein arterial signal plans are optimized with the objective of minimizing the total number of stops experienced by all vehicles on the artery.

Additionally, it is worth noting that the hyperparameters used in Yang's model and MP-BAND are derived through

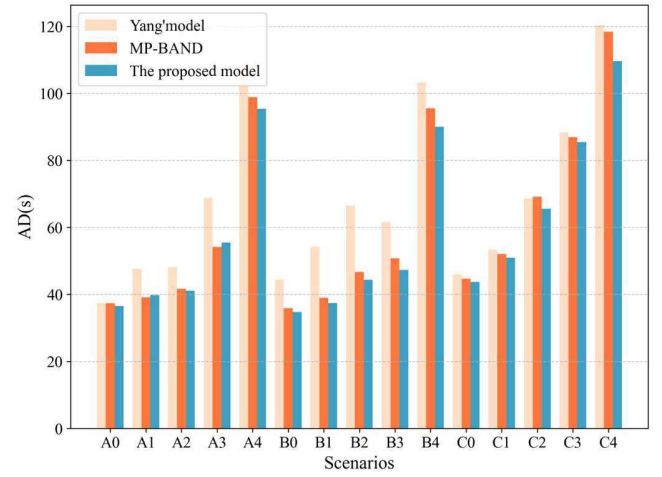


Fig. 18. Simulation results of AD(s).

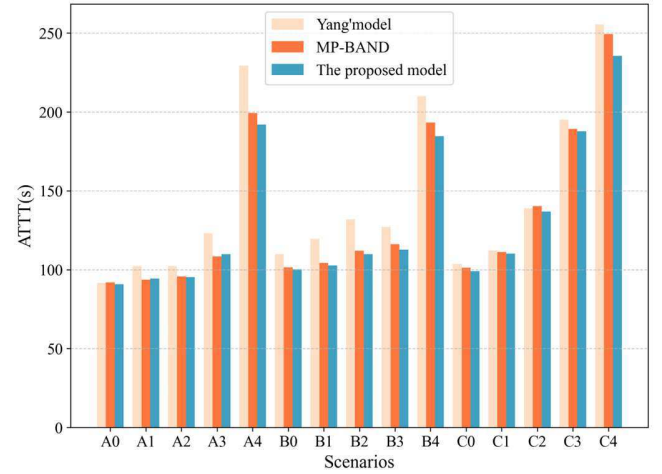


Fig. 19. Simulation results of ATTT(s).

sensitivity analysis. As a result, their optimal values are practically unattainable in real-world scenarios due to both the lack of theoretical guidance and the infeasibility of conducting extensive field experiments for parameter tuning.

#### D. Results Analysis

In order to compare the performance of different models, average delay (AD), average stops (AS) and average total travel time (ATTT) of the artery are employed as the MOEs for the experiment. The results are given as Fig. 18, Fig. 19 and Fig. 20. The improvements in MOEs by the proposed model against the other two models are also demonstrated in Fig. 21 and Fig. 22. Fig. 23 is given to show the alleviation of phase saturation by the proposed model.

As indicated by the simulation results, both MP-BAND and the proposed model outperform Yang's model in generating effective signal progression plans for arterial corridors across a range of traffic demand distributions and volume levels. Overall, the proposed model demonstrates a modest but consistent advantage over MP-BAND in improving time efficiency, particularly through reductions in average delay (AD) and

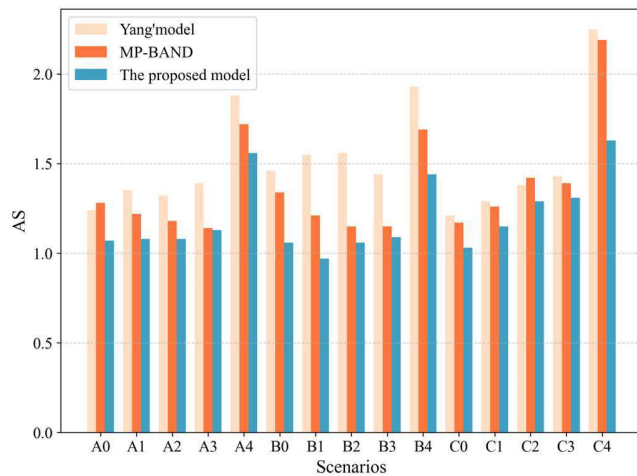


Fig. 20. Simulation results of AS(s).

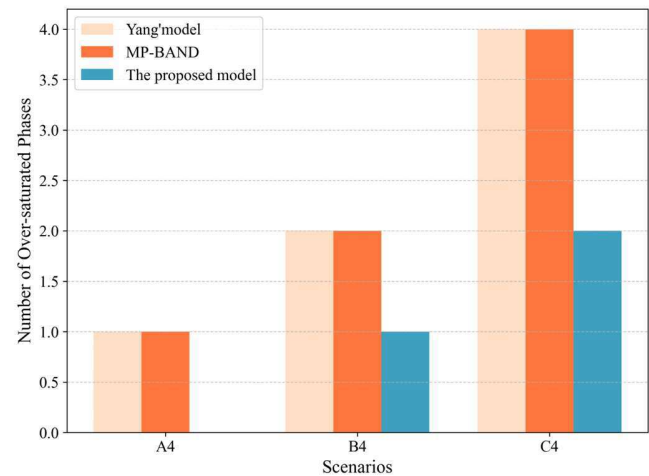


Fig. 23. A comparison of number of over-saturated phases.

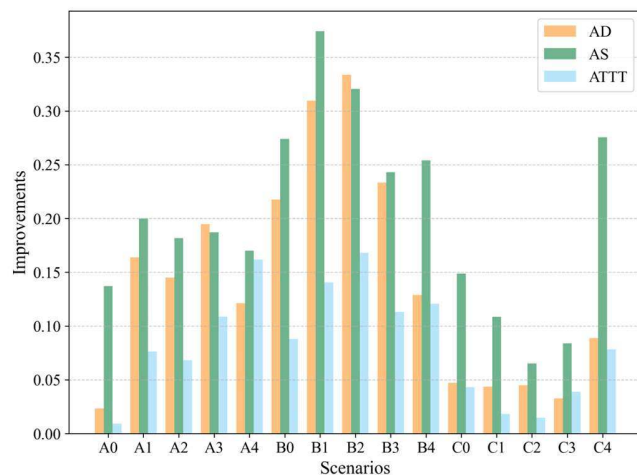


Fig. 21. Improvement in MOEs against Yang's model.

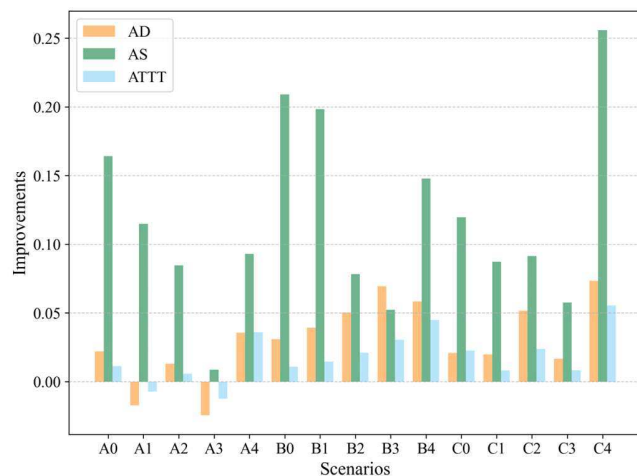


Fig. 22. Improvement in MOEs against MP-BAND.

average total travel time (ATTT). Notable exceptions include Scenarios A1 and A3, where MP-BAND achieves slightly

lower AD and ATTT values; however, the proposed model retains a clear advantage in terms of average stops (AS).

Although the improvements in AD and ATTT achieved by the proposed model are relatively modest, the reduction in vehicle stops is both substantial and consistent across all scenarios. This highlights the model's strength in optimizing system-wide efficiency. In scenarios exhibiting moderate over-saturation, the proposed model is able to adjust signal splits to mitigate system-wide saturation, thereby reducing the number of over-saturated phases, as evidenced in Fig. 23.

Nevertheless, Fig. 22 reveals a generally declining trend in the percentage of improvement in AS relative to MP-BAND as traffic volumes increase. This trend can be attributed to the diminishing effectiveness of signal progression under high saturation conditions, where vehicle movement becomes increasingly constrained. Under such conditions, the proposed model approaches the theoretical lower bound of vehicle stops more closely than MP-BAND; however, the narrowing margin for improvement at higher saturation levels results in less pronounced gains.

A similar pattern is observed in scenario group C, which focuses on optimizing two-way arterial demand. However, in Scenario C4, a marked improvement is observed, likely due to a reduction in over-saturation levels for cross-street through phases 4 and 8. Unlike Yang's model and MP-BAND, which rely on fixed, predetermined signal splits and thus cannot alleviate over-saturation, the proposed model is able to reallocate green time. This adjustment successfully shifts two of the four previously over-saturated phases into an under-saturated state. Such reallocation not only enhances operational efficiency and safety but also demonstrates the model's comprehensive approach to arterial optimization—explicitly accounting for cross-street demand, which is often overlooked in existing models.

An examination of the performance of the three models reveals that traditional approaches, such as Yang's model and MP-BAND, are inherently oriented toward improving operational efficiency, primarily due to their reliance on fixed signal splits. In contrast, the proposed model demonstrates

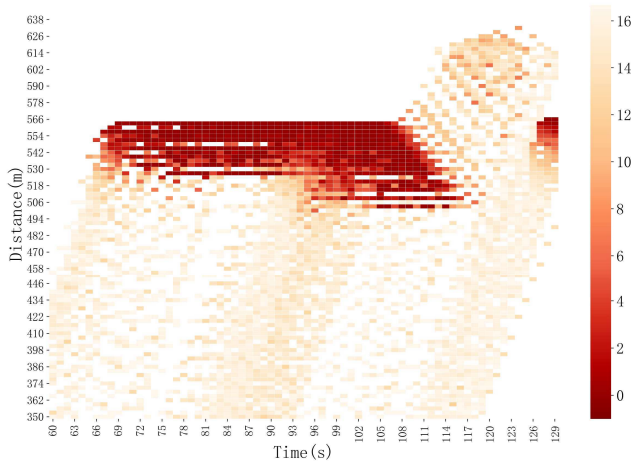


Fig. 24. Speed heat map by MP-BAND for intersection 2 in scenario A0.

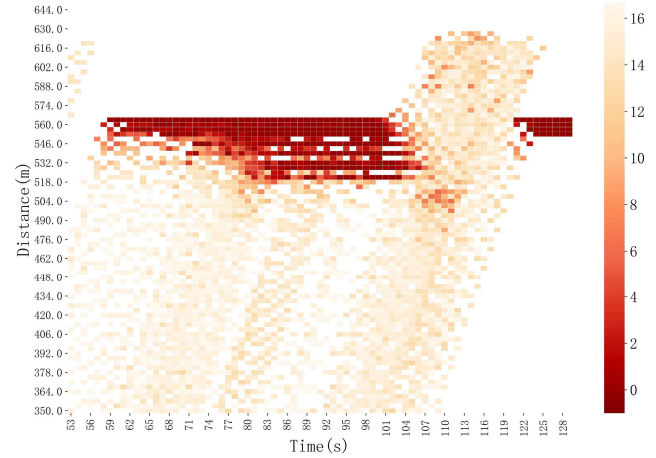


Fig. 25. Speed heat map by the proposed model for intersection 2 in scenario A0.

greater versatility by adjusting its optimization focus based on traffic conditions. Specifically, for under-saturated conditions, the model prioritizes efficiency enhancements. However, as saturation levels increase and the system approaches or exceeds capacity, the model progressively shifts its emphasis toward safety considerations—such as reducing the likelihood of spill-back and minimizing conflict points—by reallocating green time to mitigate over-saturation.

Furthermore, Scenarios B0 and B1 offer an intriguing observation in Fig. 20: while average delay typically increases with longer cycle lengths, the number of stops can still decline. This phenomenon may result from a complex interplay of factors including signal timing parameters and arterial geometry. As this interaction falls outside the core focus of the present study, it is not explored further here.

#### E. Queue Profile Comparison

Quantitative analysis of the simulation results demonstrates that the proposed model outperforms the two benchmark models in most scenarios. To further elucidate the mechanism by which the proposed model enhances arterial system efficiency, an intuitive comparison is provided based on visualized queue profiles. Fig. 24 and Fig. 25 present speed heat maps depicting vehicle movements going through the northbound through phase at intersection 2 under Scenario A0 for MP-BAND and the proposed model, respectively. The heat maps consist of 1s\*3.5m grids, where each grid's value corresponds to the average speed of vehicle trajectory points passing through it. The values were derived by aggregating trajectory data over a duration slightly exceeding one signal cycle to comprehensively capture queue formation and dissipation dynamics. Darker red hues indicate lower vehicle speeds.

There are several noteworthy findings in the figures.

First, the maximum queue length in Fig. 25 (the proposed model) is significantly shorter than that in Fig. 24 (MP-BAND), indicating fewer vehicle stops. While larger phase splits typically reduce queue lengths by increasing capacity, the observed improvement in this case was achieved without

substantial adjustments to phase splits. This demonstrates the efficacy of the proposed queue estimation mechanism.

Secondly, the reduction in queue length in the specific case is attributed to the delayed arrival of some traffic streams. The proposed model provides comparatively precise estimation on how queue profile evolves when signal parameters at adjacent intersections change. By comparing profiles with different queue lengths, the model ensures that higher-density traffic arrives within an optimal green window, thereby minimizing queue formation and enabling smoother throughput.

Thirdly, the performance improvement stems primarily from the proposed model's accurate queue estimation capability, a feature lacking in MP-BAND. Although MP-BAND similarly provides green-band for the northbound through phase, Fig. 24 reveals a significant discrepancy between the theoretical signal plan (Fig. 16) and the realized green time in simulation. This inconsistency demonstrates that MP-BAND's queue estimation errors lead to sub-optimal bandwidth implementation in practice, which is common in present studies with inaccurate queue estimation.

Finally, Fig. 25 reveals a localized low-speed zone on the lower right of the queue. This phenomenon likely results from vehicular conflicts from different origins, as shown by the overlapping TT and LT bandwidths depicted in Fig. 17. These observations highlight potential limitations in current traffic dynamics modeling and suggest opportunities for enhanced precision.

#### F. More Discussion

The uniform arrival assumption remains fundamental in fixed-time signal coordination control, particularly for band-based models, as it provides an idealized foundation for developing effective offset strategies and maximizing bandwidth along arterial corridors. Empirical evidence suggests that when aggregated over multiple signal cycles under moderately stable traffic conditions, this approximation yields sufficiently accurate results to maintain progression quality and minimize delays [15]. In the proposed model, the conventional

assumption is enhanced, when compared against present studies, by accounting for heterogeneous arrival patterns among distinct traffic streams (e.g., saturation flow versus free-flow conditions). However, within each individual traffic stream, arrivals are maintained as uniformly distributed to preserve model linearity and computational efficiency. Such assumption balances realism with practical solvability requirements.

As for the assumption of evenly distributed CVs, it may not always reflect real-world traffic conditions, where penetration rates can vary by location, time of day, and vehicle type, leading to spatially and temporally uneven distributions. Nevertheless, assuming an even distribution of CVs serves as a useful abstraction in control strategy design, providing a stable baseline to evaluate system performance and theoretical benefits. Moreover, several studies [40], [41] suggest that even with partial and uneven CV penetration, coordinated control strategies can still achieve measurable improvements in traffic performance.

Although the proposed model was originally developed for uni-modal traffic coordination, its framework exhibits inherent extensibility to multi-modal scenarios. Three viable adaptation strategies are proposed: (1) implementing bus trajectory estimation to account for transit-specific movement patterns, (2) explicitly incorporating bus stops as decision variables in the optimization objective, or (3) adopting a passenger-centric approach by minimizing average stops, which would automatically favor high-capacity transit vehicles. In addition, the model's applicability to multi-modal conditions requires consideration of vehicle composition effects, particularly regarding saturation flow estimation. As demonstrated in [42], the presence of larger vehicles (e.g., buses) increases saturation headway, consequently reducing the effective saturation flow rate. Through appropriate modifications accounting for these characteristics, the proposed framework can be effectively extended to multi-modal traffic management applications.

## V. CONCLUSION

This study presents a novel arterial signal progression model aimed at minimizing level of over-saturation and vehicular stops by providing reliable estimation on queue profiles. Unlike traditional approaches that treat phase splits and queues as fixed, the proposed model treats queue length as a variable, estimating it under varying signal plans through local progression linking adjacent intersections. Initially structured as a MIBLP, the model is transformed into a MILP using a McCormick envelope, reducing reliance on hyperparameter tuning and enhancing applicability across diverse traffic conditions.

To assess the model's performance, a simulated artery was developed in the SUMO environment, incorporating fifteen traffic demand scenarios. Computational efficiency of the model is analyzed, demonstrating that its scalability is adequate for application to scenarios involving longer arteries. Simulation results demonstrate that the proposed model consistently outperforms Yang's model across key MOEs, including average delay (AD), average stops (AS), and average total travel time (ATTT). Compared to MP-BAND, the proposed model achieves comparable or superior reductions

in AD and ATTT, and consistently yields a greater reduction in AS. Notably, the proposed model demonstrates the ability to actively mitigate over-saturated conditions by adjusting signal timings to reduce phase saturation levels—an outcome not achievable by the benchmark models, which rely on fixed splits.

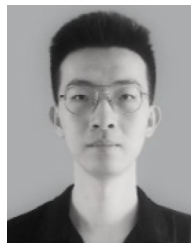
Visualizations of queue profiles using grid-based speed heat maps provide qualitative validation, demonstrating the model's accuracy in capturing queue dynamics. Discussions on the model's limitations and assumptions highlight a balanced trade-off between accuracy, computational efficiency, and effectiveness. The model also shows potential for adaptation to multi-modal traffic demands with appropriate modifications.

Future research will focus on: (1) establishing explicit connections between signal progression and additional MOEs such as delay and fuel consumption; (2) refining traffic dynamic estimations for greater accuracy; and (3) developing a framework for adaptive control in connected vehicle environments to enhance model responsiveness.

## REFERENCES

- [1] J. A. Hillier and R. Rothery, "The synchronization of traffic signals for minimum delay," *Transp. Sci.*, vol. 1, no. 2, pp. 81–94, May 1967, doi: [10.1287/trsc.1.2.81](https://doi.org/10.1287/trsc.1.2.81).
- [2] R. E. Allsop, "Selection of offsets to minimize delay to traffic in a network controlled by fixed-time signals," *Transp. Sci.*, vol. 2, no. 1, pp. 1–13, Feb. 1968, doi: [10.1287/trsc.2.1.1](https://doi.org/10.1287/trsc.2.1.1).
- [3] L.-T. Lin, L.-W. Tung, and H.-C. Ku, "Synchronized signal control model for maximizing progression along an arterial," *J. Transp. Eng.*, vol. 136, no. 8, pp. 727–735, Aug. 2010, doi: [10.1061/\(asce\)te.1943-5436.0000123](https://doi.org/10.1061/(asce)te.1943-5436.0000123).
- [4] Y. Liu and G.-L. Chang, "An arterial signal optimization model for intersections experiencing queue spillback and lane blockage," *Transp. Res. C, Emerg. Technol.*, vol. 19, no. 1, pp. 130–144, Feb. 2011, doi: [10.1016/j.trc.2010.04.005](https://doi.org/10.1016/j.trc.2010.04.005).
- [5] J. T. Morgan and J. D. C. Little, "Synchronizing traffic signals for maximal bandwidth," *Operations Res.*, vol. 12, no. 6, pp. 896–912, Dec. 1964, doi: [10.1287/opre.12.6.896](https://doi.org/10.1287/opre.12.6.896).
- [6] J. D. C. Little, "The synchronization of traffic signals by mixed-integer linear programming," *Operations Res.*, vol. 14, no. 4, pp. 568–594, Aug. 1966, doi: [10.1287/opre.14.4.568](https://doi.org/10.1287/opre.14.4.568).
- [7] J. Little, M. Kelson, and N. Gartner, "MAXBAND: A program for setting signals on arteries and triangular networks," *Transp. Res. Res. J. Transp. Res. Board*, vol. 795, pp. 40–46, Dec. 1981.
- [8] K. Lu, X. Zeng, L. Li, and J. Xu, "Two-way bandwidth maximization model with prorated impact factor for unbalanced bandwidth demands," *J. Transp. Eng.*, vol. 138, no. 5, pp. 527–534, May 2012, doi: [10.1061/\(asce\)te.1943-5436.0000352](https://doi.org/10.1061/(asce)te.1943-5436.0000352).
- [9] L. Zhang, Z. Song, X. Tang, and D. Wang, "Signal coordination models for long arterials and grid networks," *Transp. Res. C, Emerg. Technol.*, vol. 71, pp. 215–230, Oct. 2016, doi: [10.1016/j.trc.2016.07.015](https://doi.org/10.1016/j.trc.2016.07.015).
- [10] H. Zhou, H. G. Hawkins, and Y. Zhang, "Arterial signal coordination with uneven double cycling," *Transp. Res. A, Policy Pract.*, vol. 103, pp. 409–429, Sep. 2017, doi: [10.1016/j.tra.2017.07.004](https://doi.org/10.1016/j.tra.2017.07.004).
- [11] Y. Cheng, G.-L. Chang, and S. Rahwanji, "Concurrent optimization of signal progression and crossover spacing for diverging diamond interchanges," *J. Transp. Eng., A, Syst.*, vol. 144, no. 3, Mar. 2018, Art. no. 04018001, doi: [10.1061/jtepbs.0000121](https://doi.org/10.1061/jtepbs.0000121).
- [12] Y.-H. Chen, Y. Cheng, and G.-L. Chang, "Concurrent progression of through and turning movements for arterials experiencing heavy turning flows and bay-length constraints," *Transp. Res. Record, J. Transp. Res. Board*, vol. 2673, no. 9, pp. 525–537, May 2019, doi: [10.1177/0361198119843480](https://doi.org/10.1177/0361198119843480).
- [13] K. Lu, X. Tian, S. Jiang, J. Xu, and Y. Wang, "Optimization model for regional green wave coordinated control based on ring-and-barrier structure," *J. Intell. Transp. Syst.*, vol. 26, no. 1, pp. 68–80, Jan. 2022, doi: [10.1080/15472450.2020.1795847](https://doi.org/10.1080/15472450.2020.1795847).

- [14] N. H. Gartner, S. F. Assman, F. Lasaga, and D. L. Hou, "A multi-band approach to arterial traffic signal optimization," *Transp. Res. B, Methodol.*, vol. 25, no. 1, pp. 55–74, Feb. 1991, doi: [10.1016/0191-2615\(91\)90013-9](https://doi.org/10.1016/0191-2615(91)90013-9).
- [15] N. H. Gartner and C. Stamatiadis, "Arterial-based control of traffic flow in urban grid networks," *Math. Comput. Model.*, vol. 35, nos. 5–6, pp. 657–671, Mar. 2002, doi: [10.1016/s0895-7177\(02\)80027-9](https://doi.org/10.1016/s0895-7177(02)80027-9).
- [16] N. H. Gartner and C. Stamatiadis, "Progression optimization featuring arterial- and route-based priority signal networks," *J. Intell. Transp. Syst.*, vol. 8, no. 2, pp. 77–86, Apr. 2004, doi: [10.1080/15472450490437771](https://doi.org/10.1080/15472450490437771).
- [17] C. Zhang, Y. Xie, N. H. Gartner, C. Stamatiadis, and T. Arsava, "AM-band: An asymmetrical multi-band model for arterial traffic signal coordination," *Transp. Res. C, Emerg. Technol.*, vol. 58, pp. 515–531, Sep. 2015, doi: [10.1016/j.trc.2015.04.014](https://doi.org/10.1016/j.trc.2015.04.014).
- [18] J. Yao, C. Tan, and K. Tang, "An optimization model for arterial coordination control based on sampled vehicle trajectories: The STREAM model," *Transp. Res. C, Emerg. Technol.*, vol. 109, pp. 211–232, Dec. 2019, doi: [10.1016/j.trc.2019.10.014](https://doi.org/10.1016/j.trc.2019.10.014).
- [19] B. Jing, Y. Lin, Y. Shou, K. Lu, and J. Xu, "Pband: A general signal progression model with phase optimization along urban arterial," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 1, pp. 344–354, Jan. 2022, doi: [10.1109/TITS.2020.3010841](https://doi.org/10.1109/TITS.2020.3010841).
- [20] G. Dai, H. Wang, and W. Wang, "A bandwidth approach to arterial signal optimisation with bus priority," *Transportmetrica A, Transp. Sci.*, vol. 11, no. 7, pp. 579–602, Aug. 2015, doi: [10.1080/23249935.2015.1049675](https://doi.org/10.1080/23249935.2015.1049675).
- [21] Y. Lin, X. Yang, and N. Zou, "Passive transit signal priority for high transit demand: Model formulation and strategy selection," *Transp. Lett.*, vol. 11, no. 3, pp. 119–129, May 2019, doi: [10.1080/19427867.2017.1295899](https://doi.org/10.1080/19427867.2017.1295899).
- [22] W. Ma, L. Zou, K. An, N. H. Gartner, and M. Wang, "A partition-enabled multi-mode band approach to arterial traffic signal optimization," *IEEE Trans. Intell. Transp. Syst.*, vol. 20, no. 1, pp. 313–322, Jan. 2019, doi: [10.1109/TITS.2018.2815520](https://doi.org/10.1109/TITS.2018.2815520).
- [23] H. Kim, Y. Cheng, and G.-L. Chang, "Variable signal progression bands for transit vehicles under dwell time uncertainty and traffic queues," *IEEE Trans. Intell. Transp. Syst.*, vol. 20, no. 1, pp. 109–122, Jan. 2019, doi: [10.1109/TITS.2018.2801567](https://doi.org/10.1109/TITS.2018.2801567).
- [24] X. Yang, Y. Cheng, and G.-L. Chang, "A multi-path progression model for synchronization of arterial traffic signals," *Transp. Res. C, Emerg. Technol.*, vol. 53, pp. 93–111, Apr. 2015, doi: [10.1016/j.trc.2015.02.010](https://doi.org/10.1016/j.trc.2015.02.010).
- [25] T. Arsava, Y. Xie, and N. H. Gartner, "Arterial progression optimization using OD-BAND: Case study and extensions," *Transp. Res. Record, J. Transp. Res. Board*, vol. 2558, no. 1, pp. 1–10, Jan. 2016, doi: [10.3141/2558-01](https://doi.org/10.3141/2558-01).
- [26] T. Arsava, Y. Xie, and N. Gartner, "OD-NETBAND: An approach for origin–destination based network progression band optimization," *Transp. Res. Record, J. Transp. Res. Board*, vol. 2672, no. 18, pp. 58–70, Dec. 2018, doi: [10.1177/0361198118793007](https://doi.org/10.1177/0361198118793007).
- [27] H. Yan, F. He, X. Lin, J. Yu, M. Li, and Y. Wang, "Network-level multiband signal coordination scheme based on vehicle trajectory data," *Transp. Res. Part C, Emerg. Technol.*, vol. 107, pp. 266–286, Oct. 2019, doi: [10.1016/j.trc.2019.08.014](https://doi.org/10.1016/j.trc.2019.08.014).
- [28] C. Li, H. Wang, and Y. Lu, "A multi-path arterial progression model with variable signal structures," *Transportmetrica A, Transp. Sci.*, vol. 19, no. 3, pp. 1–35, Jul. 2022, doi: [10.1080/23249935.2022.2101708](https://doi.org/10.1080/23249935.2022.2101708).
- [29] C. Li, H. Wang, and Y. Lu, "Design of network green bands considering trams," *J. Transp. Eng., A, Syst.*, vol. 148, no. 12, Dec. 2022, Art. no. 04022108, doi: [10.1061/jtepbs.0000765](https://doi.org/10.1061/jtepbs.0000765).
- [30] Q. Wang, Y. Yuan, X. T. Yang, and Z. Huang, "Adaptive and multi-path progression signal control under connected vehicle environment," *Transp. Res. C, Emerg. Technol.*, vol. 124, Mar. 2021, Art. no. 102965, doi: [10.1016/j.trc.2021.102965](https://doi.org/10.1016/j.trc.2021.102965).
- [31] Y.-H. Chen, Y. Cheng, and G.-L. Chang, "Design of an arterial signal progression plan for multi-path flows with only intersection turning counts," *Transp. Res. C, Emerg. Technol.*, vol. 130, Sep. 2021, Art. no. 103322, doi: [10.1016/j.trc.2021.103322](https://doi.org/10.1016/j.trc.2021.103322).
- [32] L. Xu, J. Xu, X. Qu, and S. Jin, "An origin-destination demands-based multipath-band approach to time-varying arterial coordination," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 10, pp. 17784–17800, Oct. 2022, doi: [10.1109/TITS.2022.3150977](https://doi.org/10.1109/TITS.2022.3150977).
- [33] R. Karupiah and I. E. Grossmann, "Global optimization for the synthesis of integrated water systems in chemical processes," *Comput. Chem. Eng.*, vol. 30, no. 4, pp. 650–673, Feb. 2006, doi: [10.1016/j.compchemeng.2005.11.005](https://doi.org/10.1016/j.compchemeng.2005.11.005).
- [34] P. M. Castro, "Tightening piecewise McCormick relaxations for bilinear problems," *Comput. Chem. Eng.*, vol. 72, pp. 300–311, Jan. 2015, doi: [10.1016/j.compchemeng.2014.03.025](https://doi.org/10.1016/j.compchemeng.2014.03.025).
- [35] I. Yun and B. Park, "Stochastic optimization for coordinated actuated traffic signal systems," *J. Transp. Eng.*, vol. 138, no. 7, pp. 819–829, Jul. 2012, doi: [10.1061/\(asce\)jtr.1943-5436.0000384](https://doi.org/10.1061/(asce)jtr.1943-5436.0000384).
- [36] L. T. Truong, M. Sarvi, G. Currie, and T. M. Garoni, "How many simulation runs are required to achieve statistically confident results: A case study of simulation-based surrogate safety measures," in *Proc. IEEE 18th Int. Conf. Intell. Transp. Syst.*, Sep. 2015, pp. 274–278, doi: [10.1109/ITSC.2015.54](https://doi.org/10.1109/ITSC.2015.54).
- [37] X. Wen, D. Wang, S. Jin, G. Qian, and Y. Zhu, "A signal coordination model for long arterials considering link traffic flow characteristics," *Sustainability*, vol. 15, no. 20, p. 14874, Oct. 2023, doi: [10.3390/su152014874](https://doi.org/10.3390/su152014874).
- [38] C. Wu, Q. Cao, G. Ren, W. Chen, and Z. Zhang, "Path-based progression model for long arterial signal coordination considering partition cost," *Transp. Res. Record, J. Transp. Res. Board*, vol. 2679, no. 3, pp. 422–437, Oct. 2024, doi: [10.1177/03611981241278347](https://doi.org/10.1177/03611981241278347).
- [39] D. Machado, "A benchmark of optimization solvers for genome-scale metabolic modeling of organisms and communities," *mSystems*, vol. 9, no. 2, Feb. 2024, Art. no. 00833-23, doi: [10.1128/msystems.00833-23](https://doi.org/10.1128/msystems.00833-23).
- [40] J. Chen, L.-J. Qian, and L. Xuan, "Cooperative control of connected hybrid electric vehicles and traffic signals at isolated intersections," *IET Intell. Transp. Syst.*, vol. 14, no. 13, pp. 1903–1912, Dec. 2020, doi: [10.1049/iet-its.2020.0287](https://doi.org/10.1049/iet-its.2020.0287).
- [41] X. Xiao, Y. Zhang, X. B. Wang, S. Yang, and T. Chen, "Hierarchical longitudinal control for connected and automated vehicles in mixed traffic on a signalized arterial," *Sustainability*, vol. 13, no. 16, p. 8852, Aug. 2021, doi: [10.3390/su13168852](https://doi.org/10.3390/su13168852).
- [42] Y. Wang, J. Rong, W. Luo, and Y. Gao, "Use of an improved car-following model to explain the influence of traffic composition on saturation headway at signalized intersections," *J. Adv. Transp.*, vol. 2022, pp. 1–20, Jun. 2022, doi: [10.1155/2022/5025393](https://doi.org/10.1155/2022/5025393).



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